Requirements Specification Document – WSSOA (IT644)
Autumn 2026

Project Title :

PriceWise – Compare, Analyze, Shop Smarter

A Web-Based AI-Powered Smart Shopping Assistant

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1. PROJECT OVERVIEW

1.1 Business Context

Online shopping has significantly increased the number of products and purchasing options available to consumers. However, having more choices does not necessarily make purchasing decisions easier.

Users frequently need to visit multiple shopping platforms to compare prices, investigate product specifications, check availability, analyse previous prices, and determine whether the current price represents a good purchasing opportunity.

Traditional price comparison solutions primarily answer:

"Where can I find this product at the lowest price?"

PriceWise expands this concept by addressing a more important decision:

"Is this the right time to buy this product?"

PriceWise is proposed as a web-based AI-powered smart shopping assistant that combines product discovery, multi-source price comparison, product specification analysis, historical price intelligence, price prediction, and purchasing recommendations into a unified platform.

1.2 Problem Statement

Consumers face several challenges while making online purchasing decisions:

Fragmented Information

Product information and pricing are distributed across different shopping platforms, requiring users to manually compare multiple sources.

Lack of Historical Context

A user can see the current price but may not know whether it is high, low, or average compared with previous prices.

Difficulty Predicting Price Movement

Users often do not know whether waiting may result in a lower price or whether the current price represents a favourable purchasing opportunity.

Time-Consuming Research

Comparing products, specifications, prices, and historical trends manually can require considerable time and effort.

Decision Uncertainty

Even after collecting information, users may remain uncertain about whether to purchase immediately or wait.

PriceWise aims to address these challenges by combining these decision-making capabilities within a single system.

1.3 Business Objectives

The primary objectives of PriceWise are:

1. Provide users with centralized product discovery and comparison.
2. Reduce the time required to research products across multiple sources.
3. Enable users to compare prices from supported sources.
4. Provide historical price information for better purchasing context.
5. Visualize price trends in an understandable manner.
6. Use machine learning to estimate potential future price trends.
7. Provide actionable Buy, Wait, or Monitor recommendations.
8. Allow users to maintain a personal watchlist.
9. Allow users to configure target-price alerts.
10. Provide a centralized dashboard for monitored products and purchasing insights.
11. Create an intuitive and responsive shopping intelligence interface.

1.4 Project Scope

In Scope

The initial PriceWise system will include:

- User registration and authentication
- Product search
- Product filtering and sorting
- Product details
- Product specification display
- Multi-source price comparison
- Product comparison
- Historical price tracking
- Historical price visualization
- Price trend analysis
- AI/ML-based price prediction
- Buy/Wait/Monitor recommendation
- Watchlist
- Price alerts
- Personalized user dashboard
- Administrative product and user management

Out of Scope

The following capabilities are outside the scope of the initial PriceWise implementation:

- Direct purchase of products
- Payment processing
- Order placement
- Delivery tracking
- Seller inventory management
- Logistics management
- Guaranteed future price prediction
- Direct financial transactions
- Full e-commerce marketplace functionality

  PriceWise acts as a **shopping intelligence and decision-support platform**, rather than an online marketplace.

2. STAKEHOLDERS AND USER GROUPS

2.1 Stakeholder Identification

Stakeholder	Role	Primary Requirement
Guest User	Potential customer	Discover and compare products
Registered User	Primary customer	Track products and receive personalized insights
Administrator	System operator	Manage users, products and system information
External Data Source	Data provider	Supply product and price information
Project Team	Developers and designers	Build and maintain PriceWise
Project Supervisor	Academic reviewer	Review requirements and project progress

3. FUNCTIONAL REQUIREMENTS

Functional requirements describe **what PriceWise shall do**.

Each requirement is assigned a unique identifier so that it can be traced to user stories, use cases, and acceptance criteria.

3.1 USER AND ACCOUNT MANAGEMENT

FR-001 — User Registration

The system shall allow a new user to create a PriceWise account by providing the required registration information.

The system shall validate submitted information before creating the account.

Priority: Must Have

Primary Actor: Guest User

FR-002 — User Authentication

The system shall allow registered users to securely authenticate using their registered credentials.

The system shall validate the credentials before granting access to authenticated functionality.

Priority: Must Have

FR-003 — User Logout

The system shall allow authenticated users to terminate their active session.

After logout, protected user-specific information shall not be accessible without authentication.

Priority: Must Have

FR-004 — User Profile Management

The system shall allow registered users to view and update permitted profile information.

Priority: Should Have

FR-005 — Authentication Error Handling

The system shall provide appropriate feedback when invalid registration or authentication information is submitted without exposing sensitive security information.

Priority: Must Have

3.2 PRODUCT DISCOVERY

FR-006 — Product Search

The system shall allow users to search for products using product names, keywords, brands, or categories.

Priority: Must Have

FR-007 — Search Suggestions

The system should provide relevant search suggestions while the user enters a query.

Priority: Should Have

FR-008 — Search Result Ranking

The system shall organize search results according to relevance to the submitted query.

Priority: Should Have

FR-009 — Product Filtering

The system shall allow users to filter results using criteria such as price range, brand, category, rating, availability, and relevant product attributes.

Priority: Must Have

FR-010 — Product Sorting

The system shall allow users to sort search results according to supported criteria such as price, relevance, and rating.

Priority: Should Have

3.3 PRODUCT INFORMATION

FR-011 — Product Details

The system shall display detailed information for a selected product, including product name, brand, category, image, description, specifications, current price, source, and availability where available.

Priority: Must Have

FR-012 — Product Specification Display

The system shall organize product specifications into a structured and readable format.

Priority: Must Have

FR-013 — Product Source Attribution

The system shall identify the source associated with displayed product and pricing information where applicable.

Priority: Must Have

FR-014 — Product Availability

The system shall display product availability information when such information is provided by supported data sources.

Priority: Should Have

3.4 PRICE COMPARISON

FR-015 — Multi-Source Price Comparison

The system shall retrieve and display available prices for a product from supported sources.

Each price shall be associated with its corresponding source.

Priority: Must Have

FR-016 — Lowest Available Price Identification

The system shall identify the lowest currently available price among supported sources when sufficient pricing information exists.

Priority: Must Have

FR-017 — Price Difference Analysis

The system shall allow users to identify differences between available prices for the same product.

The system may display both absolute and percentage price differences.

Priority: Should Have

FR-018 — Price Timestamp

The system shall associate price information with a timestamp indicating when the price was recorded or retrieved.

Priority: Must Have

FR-019 — Price Data Refresh

The system shall update price information according to the defined data collection or refresh mechanism.

Priority: Must Have

3.5 PRODUCT COMPARISON

FR-020 — Product Selection for Comparison

The system shall allow users to select multiple products for comparison.

Priority: Must Have

FR-021 — Side-by-Side Comparison

The system shall display selected products in a side-by-side comparison format.

Priority: Must Have

FR-022 — Specification Difference Identification

The system should assist users in identifying differences between selected product specifications.

Priority: Should Have

FR-023 — Comparison Price Analysis

The system shall display current pricing information for each selected product within the comparison interface.

Priority: Must Have

3.6 HISTORICAL PRICE INTELLIGENCE

FR-024 — Historical Price Storage

The system shall maintain historical price records for supported products.

Each record shall contain relevant product, price, source, and timestamp information where available.

Priority: Must Have

FR-025 — Price History Retrieval

The system shall allow users to access historical price information for supported products.

Priority: Must Have

FR-026 — Historical Price Visualization

The system shall display historical pricing information using graphical visualizations.

Priority: Must Have

FR-027 — Historical Price Statistics

The system should provide statistics such as:

- Lowest recorded price
- Highest recorded price
- Average price
- Current price
- Historical price range

Priority: Should Have

FR-028 — Price Trend Identification

The system shall analyse available historical information to identify general upward, downward, or stable price behaviour.

Priority: Should Have

3.7 AI/ML PRICE PREDICTION

FR-029 — Prediction Data Preparation

The system shall prepare relevant historical pricing information for use by the prediction component.

This may include validation, cleaning, transformation, and feature preparation.

Priority: Must Have

FR-030 — Price Prediction

The system shall use an appropriate machine learning approach to generate an estimated future price or price trend where sufficient data exists.

Priority: Should Have

FR-031 — Prediction Horizon

The prediction component shall operate over a defined prediction period established during system design and model evaluation.

Priority: Should Have

FR-032 — Prediction Confidence Indicator

Where technically supported, the system should provide an indication of prediction confidence or reliability.

Priority: Could Have

FR-033 — Prediction Data Sufficiency

The system shall determine whether sufficient historical information exists before generating a prediction.

If insufficient data exists, the system shall communicate that prediction is unavailable or unreliable.

Priority: Must Have

FR-034 — Prediction Disclaimer

The system shall clearly communicate that ML-generated predictions are estimates and do not guarantee future prices.

Priority: Must Have

3.8 PURCHASE RECOMMENDATION

FR-035 — Recommendation Generation

The system shall generate a purchase recommendation using relevant current pricing, historical pricing, price trends, and prediction results where available.

Priority: Should Have

FR-036 — Buy Recommendation

The system may classify a product as **BUY NOW** when the defined recommendation logic indicates that the current purchasing opportunity is favourable.

Priority: Should Have

FR-037 — Wait Recommendation

The system may classify a product as **WAIT** when available evidence indicates that delaying the purchase may provide a potentially better opportunity.

Priority: Should Have

FR-038 — Monitor Recommendation

The system may classify a product as **MONITOR** when available evidence does not strongly support either immediate purchase or waiting.

Priority: Should Have

FR-039 — Recommendation Explanation

The system should provide supporting information explaining the primary factors contributing to the recommendation.

These may include current price position, historical price behaviour, recent trends, and predicted movement.

Priority: Should Have

3.9 WATCHLIST AND MONITORING

FR-040 — Add Product to Watchlist

Authenticated users shall be able to add products to their personal watchlist.

Priority: Should Have

FR-041 — Remove Product from Watchlist

Authenticated users shall be able to remove products from their watchlist.

Priority: Should Have

FR-042 — View Watchlist

Authenticated users shall be able to view saved products and associated price information.

Priority: Should Have

FR-043 — Watchlist Price Monitoring

The system shall monitor relevant pricing information for products saved by users where current data is available.

Priority: Should Have

3.10 PRICE ALERTS

FR-044 — Configure Target Price

Users shall be able to specify a target price for a product they are monitoring.

Priority: Could Have

FR-045 — Trigger Price Alert

The system shall generate an alert when a monitored product satisfies the configured price condition.

Priority: Could Have

FR-046 — Manage Price Alerts

Users shall be able to view, modify, disable, and remove configured price alerts.

Priority: Could Have

3.11 USER DASHBOARD

FR-047 — Personalized Dashboard

The system shall provide authenticated users with a centralized dashboard containing relevant personalized information.

This may include:

- Watchlist
- Active alerts
- Tracked products
- Recent price changes
- Recommendations

Priority: Should Have

FR-048 — Recent Activity

The system should display relevant recent activity associated with monitored products.

Priority: Could Have

3.12 ADMINISTRATION

FR-049 — Product Data Management

Authorized administrators shall be able to manage product information maintained by the system.

Priority: Must Have

FR-050 — User Management

Authorized administrators shall be able to view and manage user accounts according to defined permissions.

Priority: Must Have

FR-051 — Data Quality Monitoring

The system should provide administrators with visibility into data quality issues such as incomplete product information or missing price records.

Priority: Should Have

FR-052 — System Monitoring

The administrative interface should provide information required to monitor important system operations and failures.

Priority: Should Have

4. NON-FUNCTIONAL REQUIREMENTS

Non-functional requirements define the quality, security, reliability, usability, and operational characteristics of PriceWise.

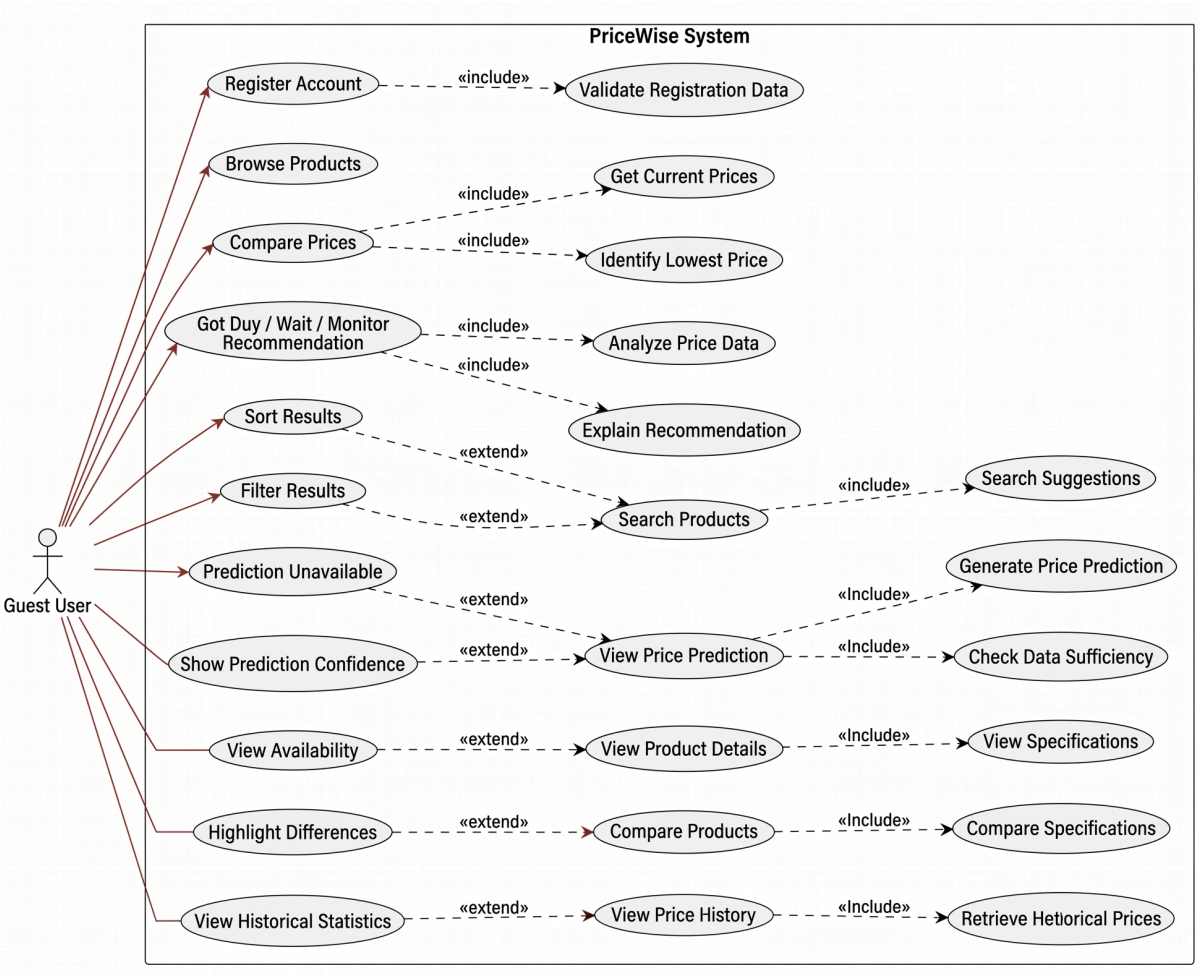
ID	Category	Requirement	Priority
NFR-001	Performance	The system shall provide responsive interaction for normal user operations under the expected project workload.	Must Have
NFR-002	Scalability	The architecture shall support growth in users, products, and historical price records.	Should Have
NFR-003	Availability	The platform should remain available during defined operating periods except during planned maintenance or unavoidable service interruptions.	Must Have
NFR-004	Security	User credentials and sensitive information shall be protected using appropriate security mechanisms.	Must Have
NFR-005	Authentication Security	Passwords shall not be stored in plain text.	Must Have
NFR-006	Authorization	Administrative functionality shall only be accessible to authorized users.	Must Have
NFR-007	Data Integrity	The system shall maintain consistency between products, prices, sources, and historical records.	Must Have
NFR-008	Data Freshness	Pricing information should provide an indication of freshness through timestamps or equivalent information.	Must Have
NFR-009	Usability	The interface shall be intuitive and understandable for users with basic technical knowledge.	Must Have

NFR-010	Accessibility	The interface should follow appropriate accessibility practices for navigation, readability, controls, and visual information.	Should Have
NFR-011	Responsiveness	The application shall support responsive layouts across desktop, tablet, and mobile screen sizes.	Must Have
NFR-012	Browser Compatibility	The system shall support commonly used modern web browsers.	Must Have
NFR-013	Maintainability	The software shall use modular components to simplify maintenance, debugging, and future enhancement.	Should Have
NFR-014	Reliability	The system shall handle unexpected failures gracefully.	Must Have
NFR-015	Error Handling	User-facing error messages shall be meaningful without exposing internal implementation details.	Must Have
NFR-016	Prediction Transparency	The system shall clearly distinguish actual historical prices from ML-generated estimates.	Must Have
NFR-017	Privacy	The system shall limit collection and storage of personal information to information required for intended functionality.	Must Have
NFR-018	Auditability	Important administrative and system events should be recorded where required for troubleshooting and accountability.	Should Have

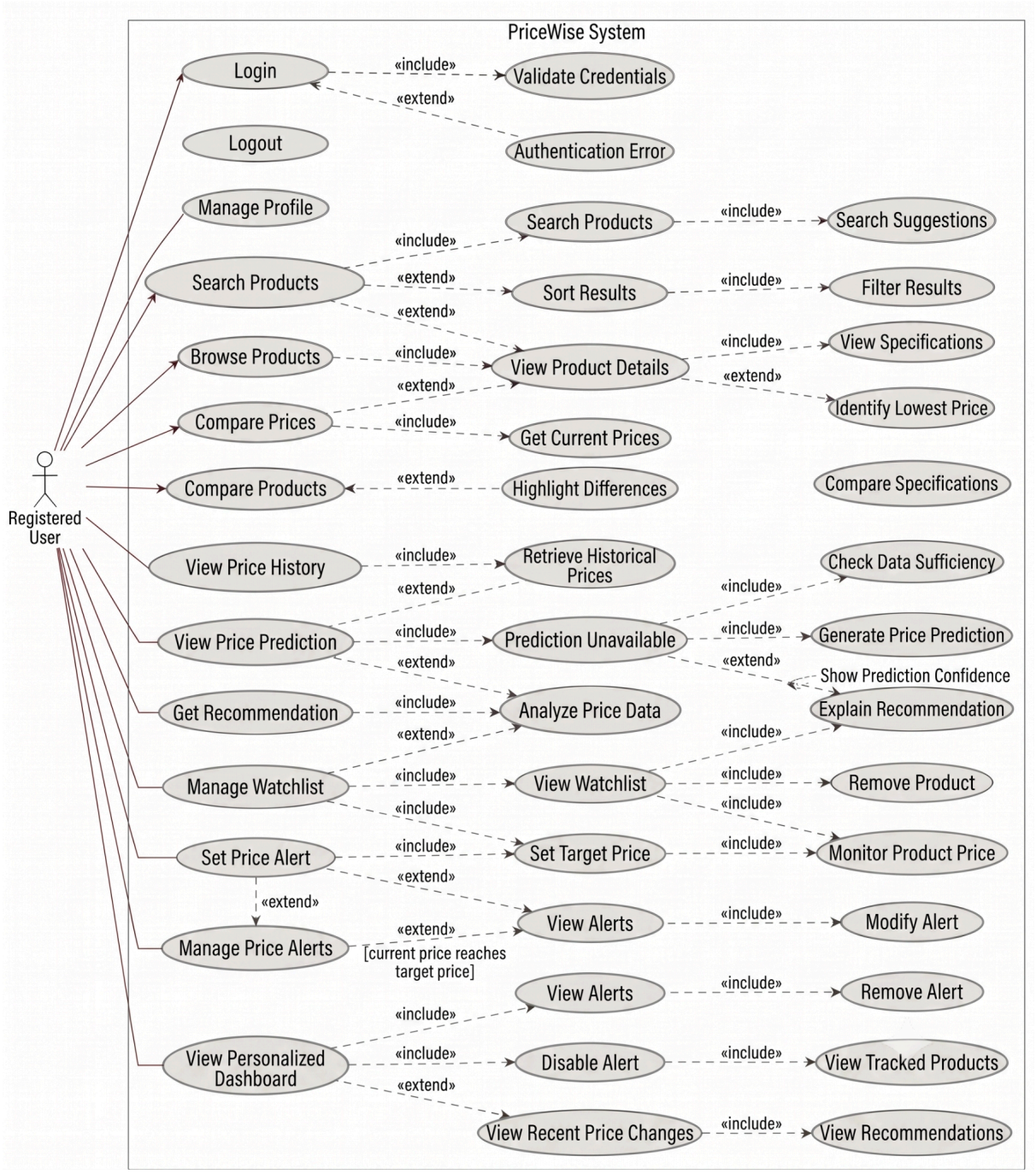
5. Use Case & User Stories

5.1 Use Case

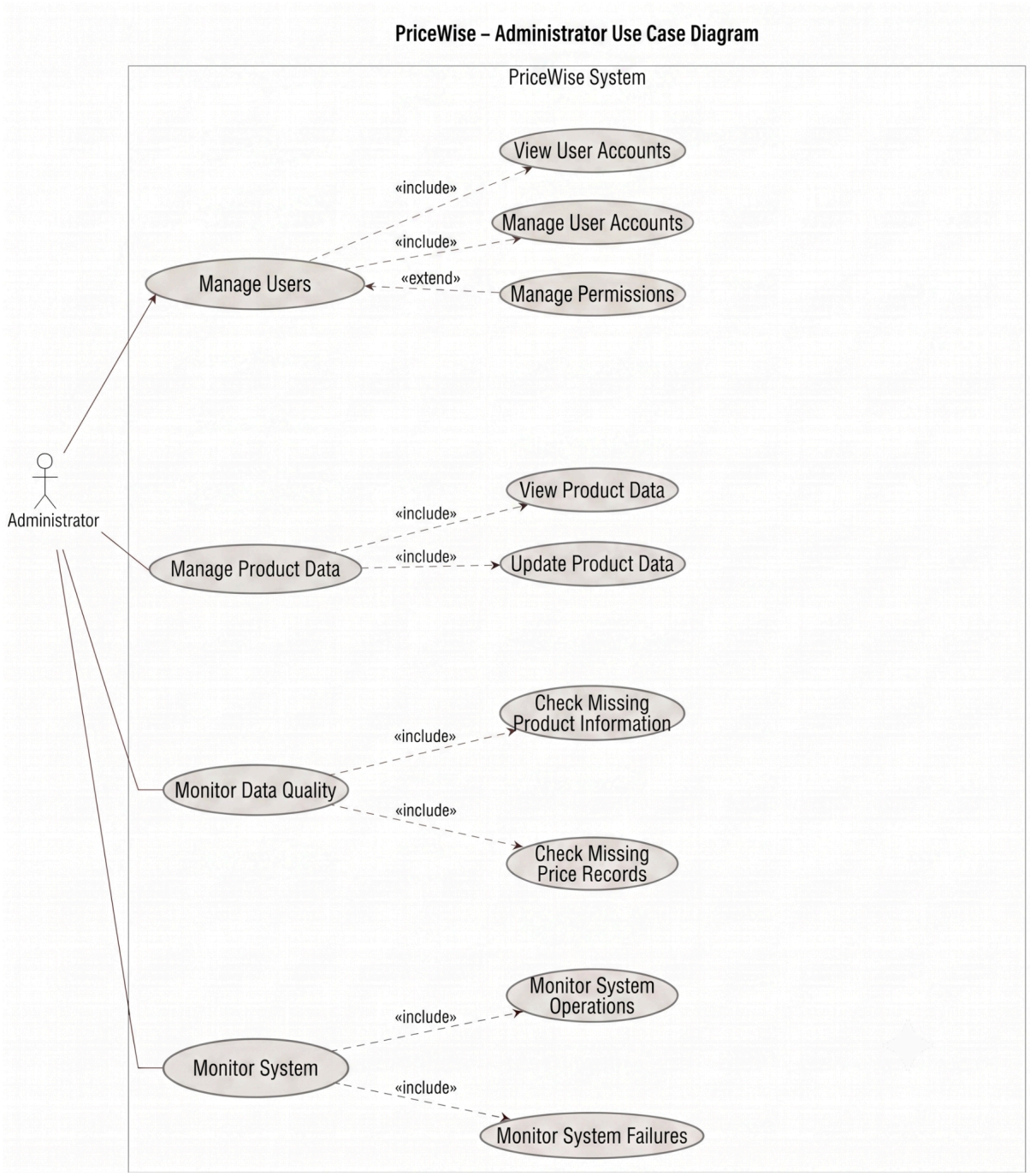
5.1.1 Guest User



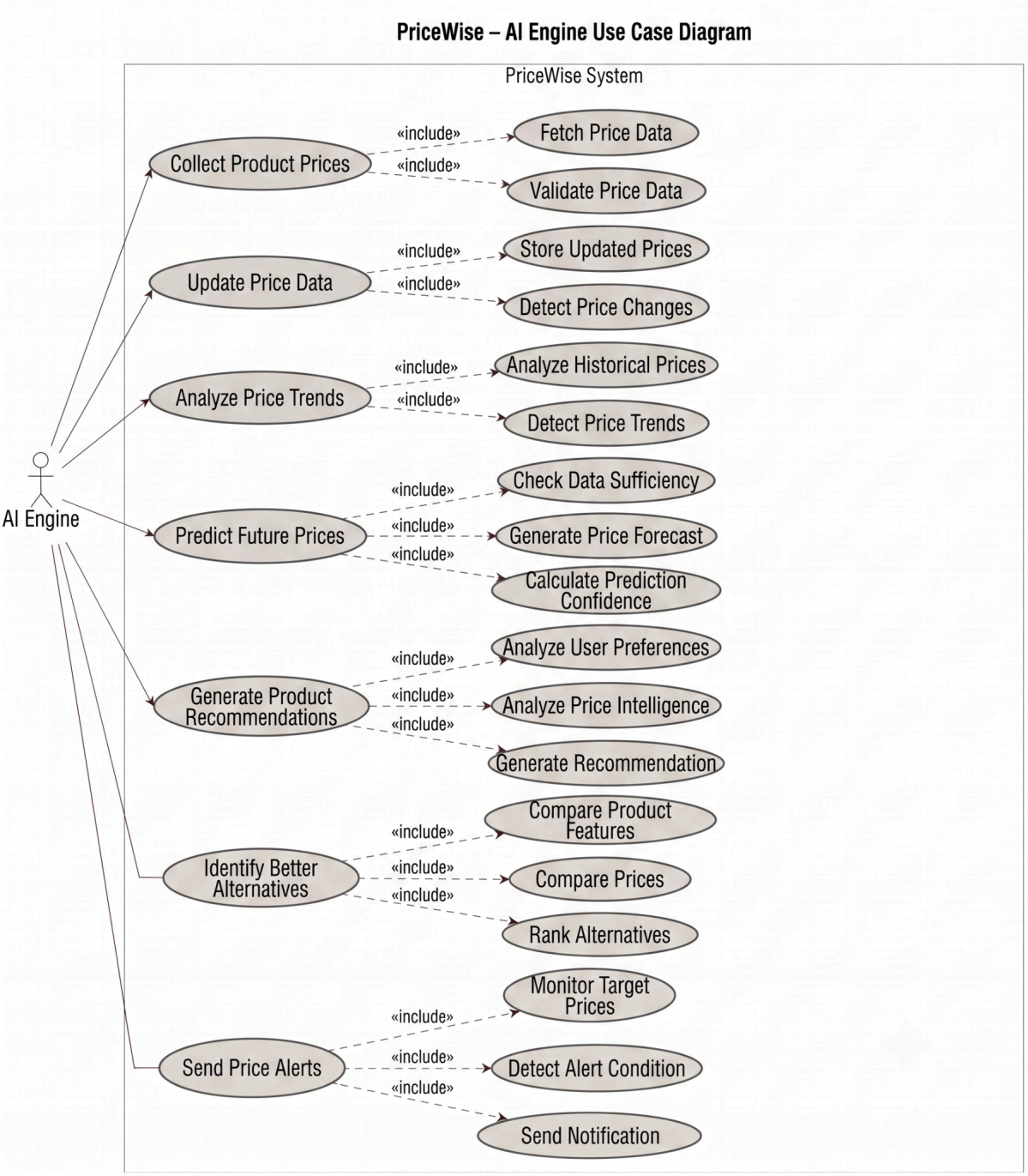
5.1.2 Registered User



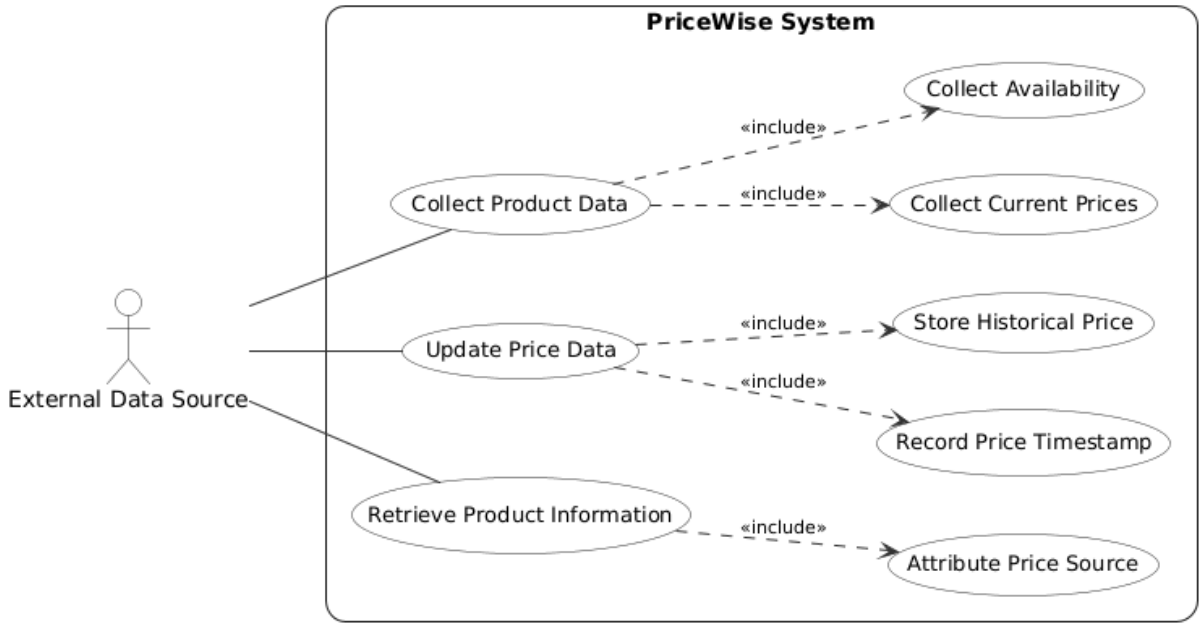
5.1.3 Admin


PriceWise – Administrator Use Case Diagram
PriceWise System
View User Accounts
«include»
Manage User Accounts
«include»
Manage Users
«extend»
Manage Permissions
Administrator
View Product Data
«include»
Manage Product Data
«include»
Update Product Data
Check Missing Product Information
«include»
Monitor Data Quality
«include»
Check Missing Price Records
Monitor System Operations
«include»
Monitor System
«include»
Monitor System Failures

5.1.4 AI Engine



5.1.5 External Data Source



5.2 USER STORIES

User stories express requirements from the perspective of the people interacting with PriceWise.

5.2.1 ACCOUNT MANAGEMENT

US-001 — Register Account

As a new user,
I want to create a PriceWise account,
so that I can access personalized shopping features.

US-002 — Login

As a registered user,
I want to securely log into my account,
so that I can access my saved products and alerts.

US-003 — Manage Profile

As a registered user,
I want to manage my profile and preferences,
so that my PriceWise experience can be personalized.

5.2.2 PRODUCT DISCOVERY

US-004 — Search Products

As a shopper,
I want to search for products by name or keyword,
so that I can quickly find products that interest me.

US-005 — Search Suggestions

As a shopper,
I want to receive search suggestions while typing,
so that I can find relevant products faster.

US-006 — Filter Results

As a shopper,
I want to filter products by price, brand, category, and other attributes,
so that I can narrow down my options.

US-007 — Sort Results

As a shopper,
I want to sort products according to price, relevance, or other criteria,
so that I can prioritize products that best match my preferences.

5.2.3 PRODUCT ANALYSIS

US-008 — View Product Details

As a shopper,
I want to view detailed product information,
so that I can understand the product before making a decision.

US-009 — Compare Prices

As a shopper,
I want to compare prices from different sources,
so that I can identify the best available deal.

US-010 — Compare Products

As a shopper,
I want to compare multiple products side-by-side,
so that I can understand their differences before purchasing.

US-011 — Compare Specifications

As a shopper,
I want to compare product specifications,
so that I can select the product that best meets my requirements.

5.2.4 PRICE INTELLIGENCE

US-012 — View Price History

As a shopper,
I want to see historical prices,
so that I can determine whether the current price is relatively high or low.

US-013 — View Price Trend

As a shopper,
I want to visualize price movement over time,
so that I can understand the product's pricing behaviour.

US-014 — View Historical Statistics

As a shopper,
I want to see historical high, low, and average prices,
so that I can put the current price into context.

5.2.5 AI/ML INTELLIGENCE

US-015 — View Predicted Price

As a shopper,
I want to see an estimated future price trend,
so that I can decide whether purchasing now may be beneficial.

US-016 — Understand Prediction Reliability

As a shopper,
I want to understand the reliability of a prediction,
so that I can make decisions with appropriate expectations.

US-017 — Receive Recommendation

As a shopper,
I want PriceWise to recommend Buy, Wait, or Monitor,
so that I can make a purchasing decision more confidently.

US-018 — Understand Recommendation

As a shopper,
I want to see the factors behind the recommendation,
so that I understand why PriceWise suggested a particular action.

5.2.6 PRODUCT MONITORING

US-019 — Add to Watchlist

As a registered user,
I want to save products to my watchlist,
so that I can monitor them later.

US-020 — Set Target Price

As a registered user,
I want to set a target price for a product,
so that I can wait for my preferred purchasing price.

US-021 — Receive Price Alert

As a registered user,
I want to receive an alert when a tracked product reaches my target price,
so that I do not need to manually check it.

US-022 — Manage Alerts

As a registered user,
I want to modify or disable my price alerts,
so that I can control which products I continue monitoring.

5.2.7 DASHBOARD

US-023 — Personalized Dashboard

As a registered user,
I want a dashboard showing my tracked products, alerts, and recommendations,
so that I can manage my shopping activity from one place.

US-024 — Monitor Price Changes

As a registered user,
I want to see recent price changes for my tracked products,
so that I can identify potential purchasing opportunities.

6. Wireframes - UI/UX mockup

This section documents each screen of the PriceWise platform. For every screen, we present:
1. Screen Purpose & User Goals
2. Low-Fidelity Blueprint Wireframe (Structural Layout with Image & Field Placeholders)
3. High-Fidelity UI/UX Figma Mockup (Final Production Visual Design)
4. Component & Layout Breakdown Table
5. Screen-Specific Acceptance Criteria

6.1 : Landing Page & Smart Home Dashboard

Route: / (Home)

Screen Purpose & User Goal: The primary entry point of the platform designed to communicate the core value proposition, facilitate instant multi-store product searches, showcase live price monitoring preview widgets, and present macro platform trust metrics.

A. Low-Fidelity Blueprint Wireframe (Structural Layout & Placeholders):

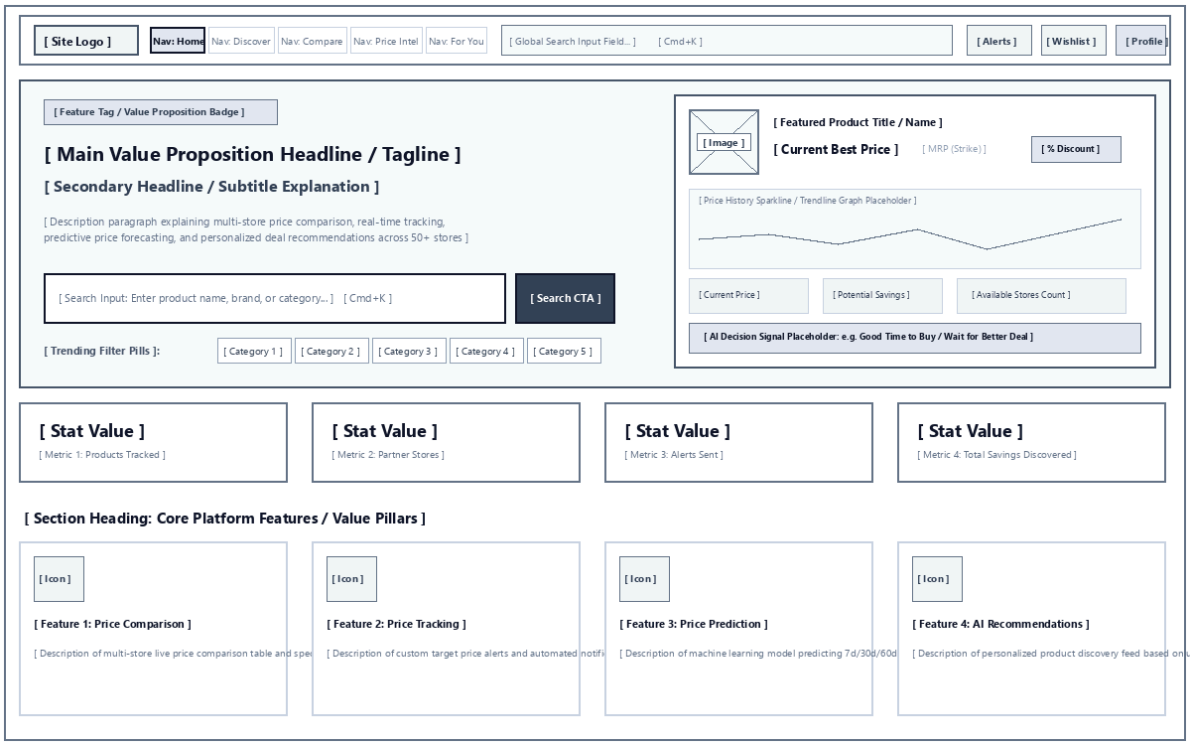


Figure 6.1(a): Low-Fidelity Blueprint Wireframe — Landing Page & Smart Home Dashboard

B. High-Fidelity UI/UX Figma Mockup :

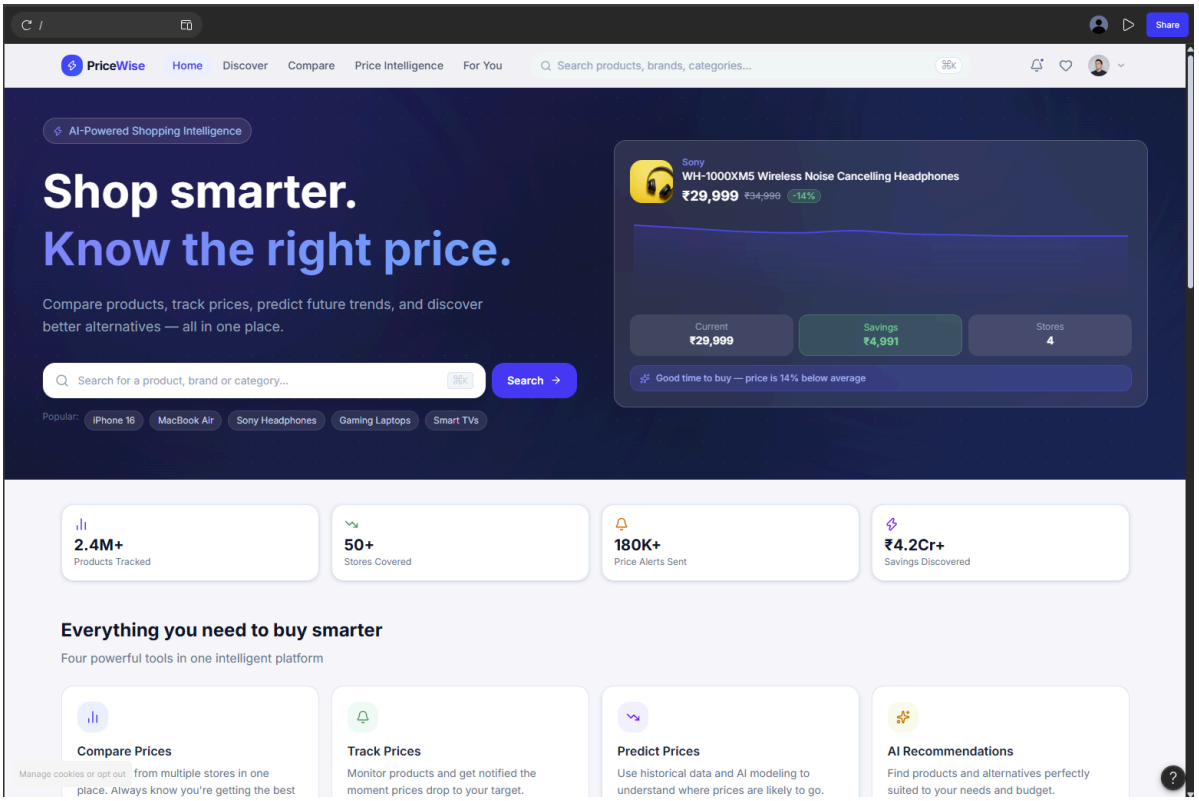


Figure 6.1(b): High-Fidelity Figma UI Mockup — Landing Page & Smart Home Dashboard

C. Interface Component & Layout Breakdown:

UI Component / Region	Visual Design & Functional Specification
Global Navigation Bar	Site logo placeholder, 5 main section navigation links with active state indicator, global search input field with Cmd+K shortcut, and right-aligned user actions (Alerts, Wishlist, Profile).
Hero Search Section	Platform tagline and value proposition headline, explanatory subtitle paragraph, primary search input field with 'Search' CTA button, and quick-filter category pills.
Live Interactive Preview Card	Featured product card containing crossed-box image placeholder, model title, current lowest price vs original MRP, discount badge, 90-day price sparkline trend graph, and AI buy/wait recommendation pill.
Platform Trust Metric Cards	Row of 4 macro statistic cards: Total Products Tracked, Partner Stores Covered, Total Alerts Sent, and Cumulative Savings Discovered.
Core Feature Value Grid	4-column feature pillar cards: Price Comparison, Real-Time Price Tracking, Predictive Price Forecasting, and Personalized AI Recommendations.

D. Screen-Specific Acceptance Criteria:

- Global search input must auto-focus upon clicking Cmd+K shortcut or clicking within the field.
- Trending category pills must redirect to `/discover` with query parameters pre-filtered.
- Featured preview card must dynamically render lowest price available across all crawled retail merchants.
- All primary navigation links must route correctly with active indicator underline on current page.

6.2: User Authentication & Sign-In Page

Route: /login (Authentication)

Screen Purpose & User Goal: Provides a clean, frictionless split-screen login and onboarding experience that re-emphasizes key platform benefits while enabling email/password sign-in and Google Single Sign-On (SSO).

A. Low-Fidelity Blueprint Wireframe (Structural Layout & Placeholders):

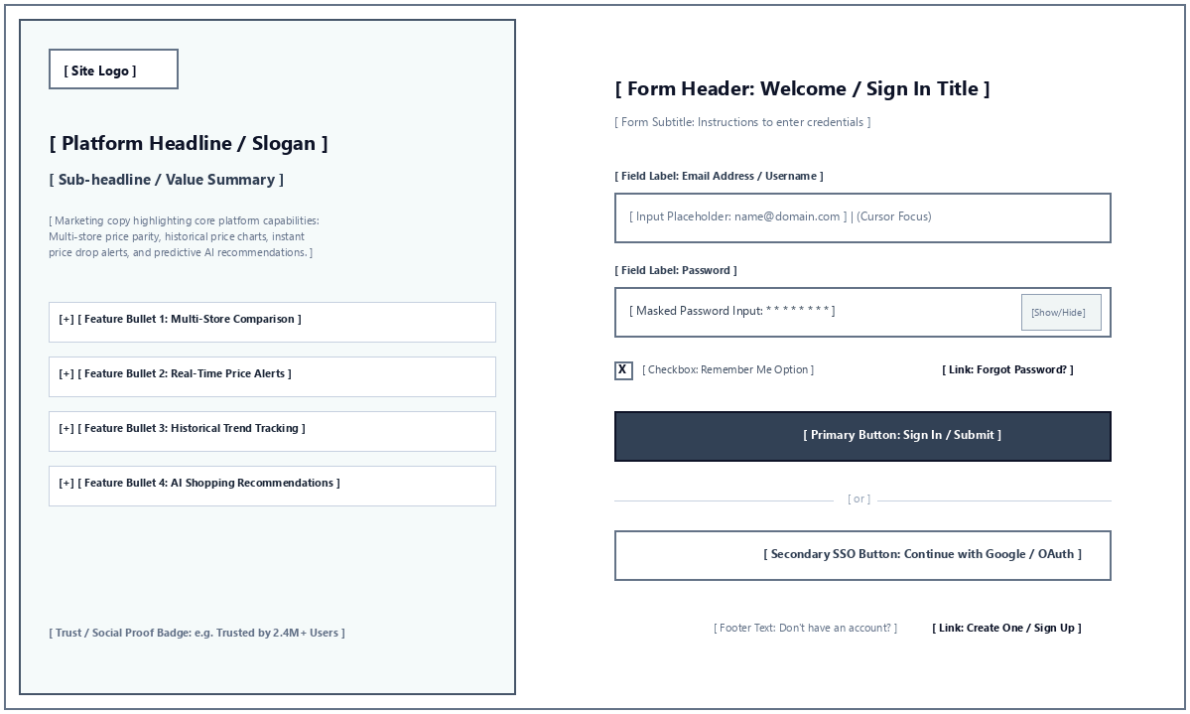


Figure 6.2(a): Low-Fidelity Blueprint Wireframe — User Authentication & Sign-In Page

B. High-Fidelity UI/UX Figma Mockup :

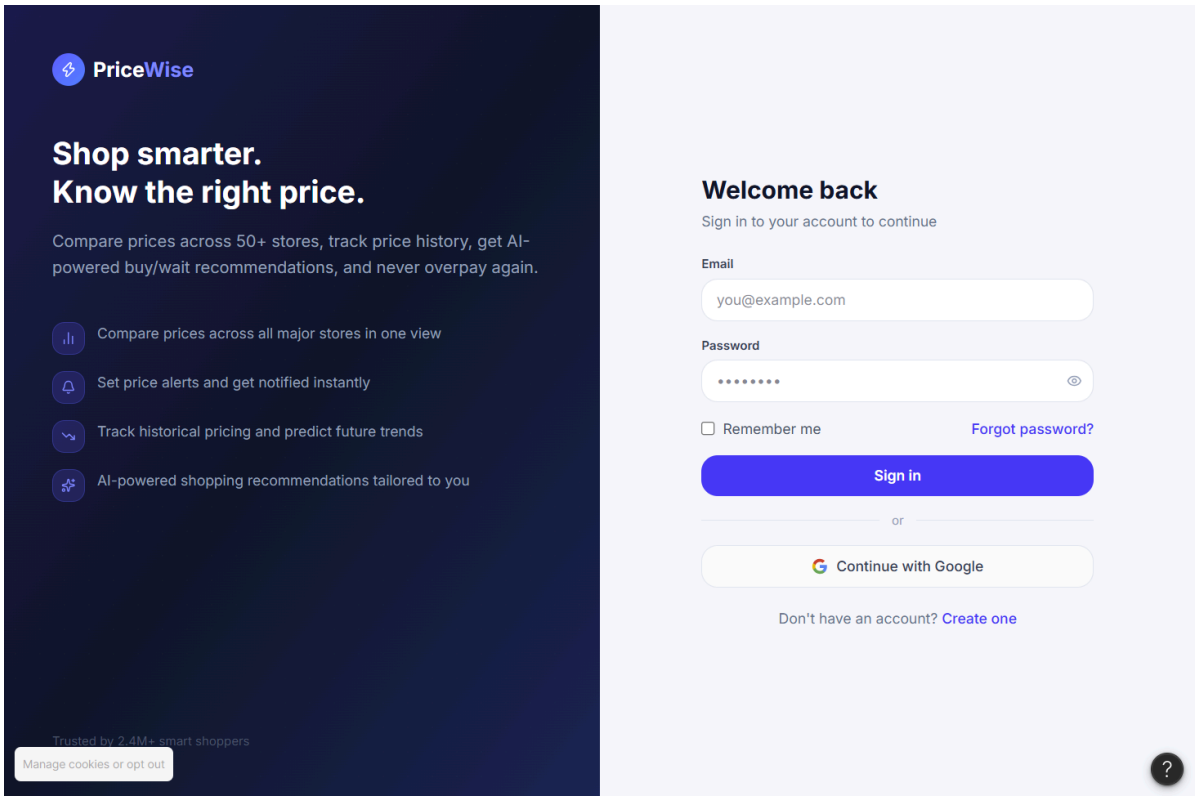


Figure 6.2(b): High-Fidelity Figma UI Mockup — User Authentication & Sign-In Page

C. Interface Component & Layout Breakdown:

UI Component / Region	Visual Design & Functional Specification
Left Value-Proposition Panel	Site logo box, prominent headline, marketing summary text, 4 feature benefit bullet blocks with icons, and social proof trust badge.
Right Authentication Form	Form title ('Welcome back'), instructional subtitle, email address text input field, password text input field with show/hide password toggle button.
Form Controls & Links	'Remember me' persistent session checkbox, 'Forgot password?' recovery text link.
Action CTAs	Primary 'Sign In' submission button, 'or' divider line, and branded 'Continue with Google' OAuth SSO button.
Registration Redirect	Footer link 'Don't have an account? Create one' routing to registration flow.

D. Screen-Specific Acceptance Criteria:

- Form validates email syntax and enforces minimum password length prior to submission.
- Password show/hide toggle switches input field type between masked and visible text.
- Google SSO initiates OAuth 2.0 authentication flow and redirects user to dashboard upon success.
- 'Remember me' checkbox persists session token in encrypted storage.

6.3: Product Discovery & Catalog Hub

Route: /discover (Explore Products)

Screen Purpose & User Goal: Enables users to explore multi-category catalogs, view trending products, filter by discount percentages, inspect best-store pricing tags, and receive instant AI buy/wait signals before adding to comparison or tracking.

A. Low-Fidelity Blueprint Wireframe (Structural Layout & Placeholders):

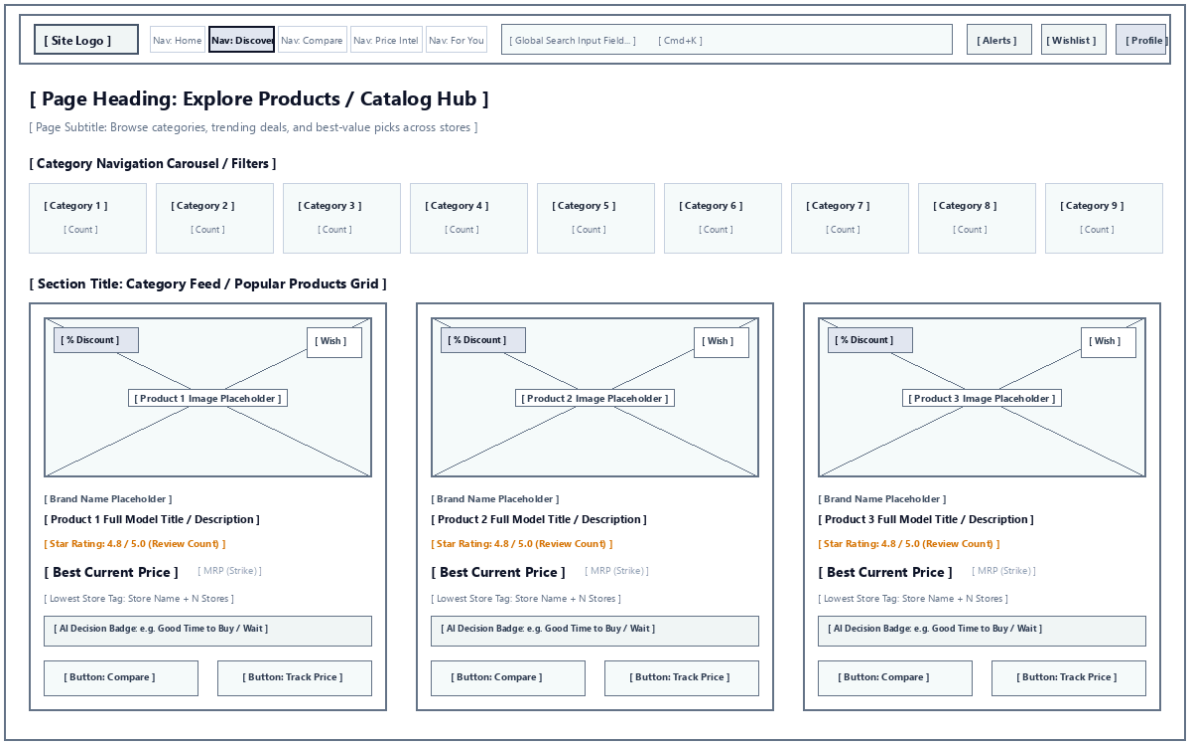


Figure 6.3(a): Low-Fidelity Blueprint Wireframe — Product Discovery & Catalog Hub

B. High-Fidelity UI/UX Figma Mockup :

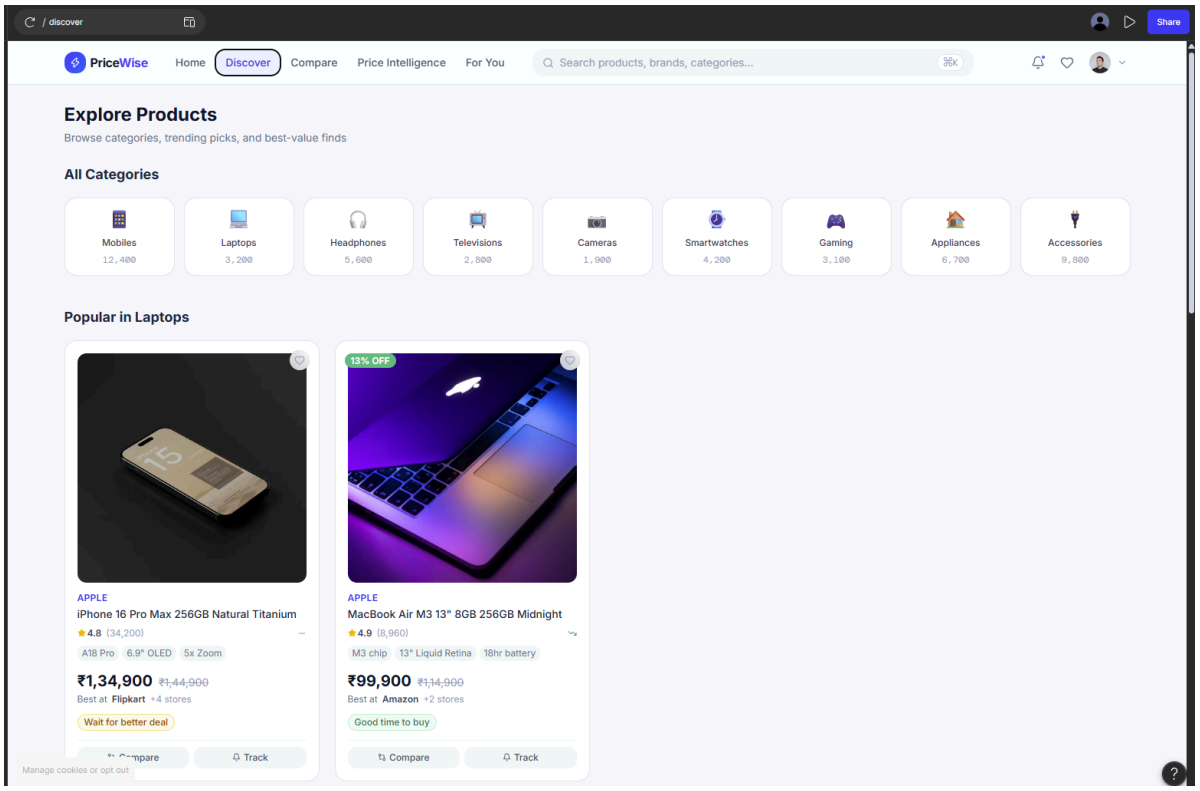


Figure 6.3(b): High-Fidelity Figma UI Mockup — Product Discovery & Catalog Hub

C. Interface Component & Layout Breakdown:

UI Component / Region	Visual Design & Functional Specification
Category Navigation Carousel	Horizontal interactive category pills with item count badges (Mobiles, Laptops, Audio, TVs, Cameras, Smartwatches, Gaming, Appliances, Accessories).
Product Cards Grid	3-column responsive product card grid containing crossed-box image placeholders, percentage discount tag, favorite/wishlist button, brand label, product model title, star rating, and hardware spec tags.
Price & Merchant Badges	Current lowest price, original MRP with strikethrough, lowest store merchant tag (e.g. Best at Flipkart +4 stores), and AI recommendation status badge ('Good time to buy' vs 'Wait for better deal').
Card Action Buttons	Dual CTA buttons: 'Compare' (adds item to comparison tray) and 'Track Price' (opens target price dialog).

D. Screen-Specific Acceptance Criteria:

- Clicking any category tile instantly filters product catalog without requiring a page reload.
- Hovering over product cards triggers subtle elevation and image zoom effects.
- Clicking 'Compare' button increments comparison tray badge count in the navigation bar.
- Clicking 'Track' opens modal dialog to configure custom price drop alerts for that product.
-

6.4: Side-by-Side Product Comparison Matrix

Route: /compare (Product Comparison)

Screen Purpose & User Goal: Offers an in-depth, multi-column comparison table allowing buyers to evaluate 2 to 4 competing products simultaneously across pricing, merchant availability, user sentiment ratings, and technical specifications.

A. Low-Fidelity Blueprint Wireframe (Structural Layout & Placeholders):

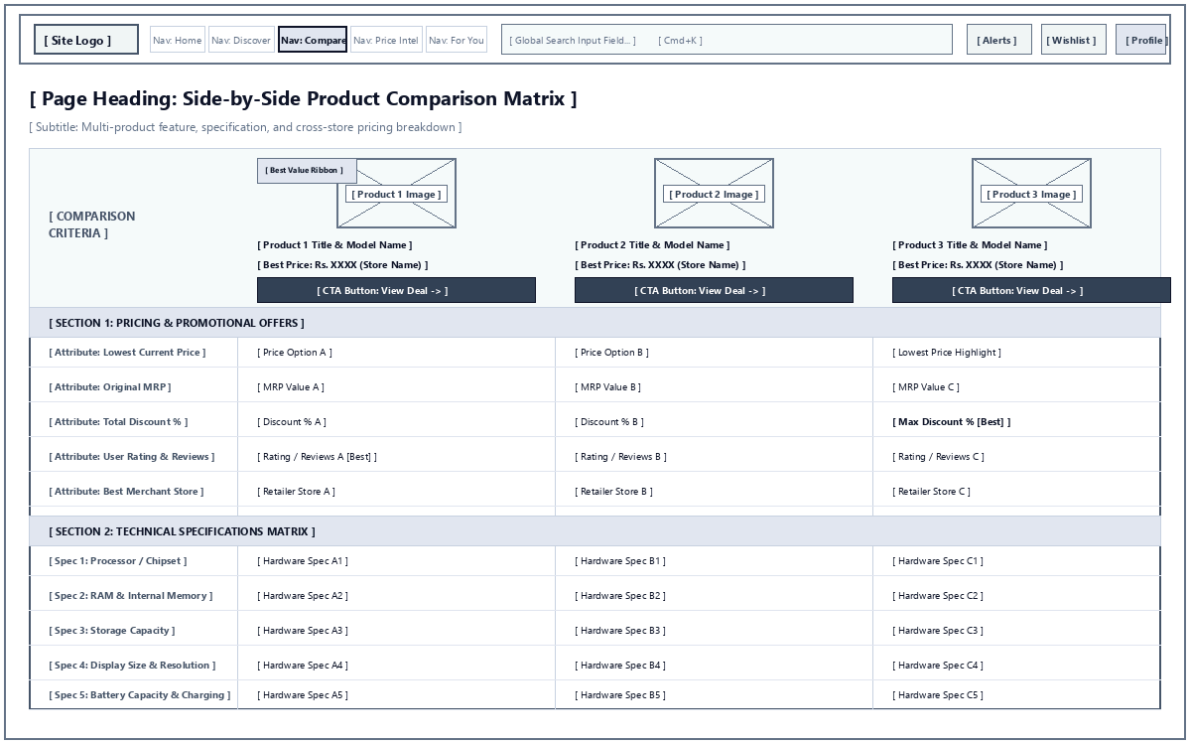


Figure 6.4(a): Low-Fidelity Blueprint Wireframe — Side-by-Side Product Comparison Matrix

B. High-Fidelity UI/UX Figma Mockup :

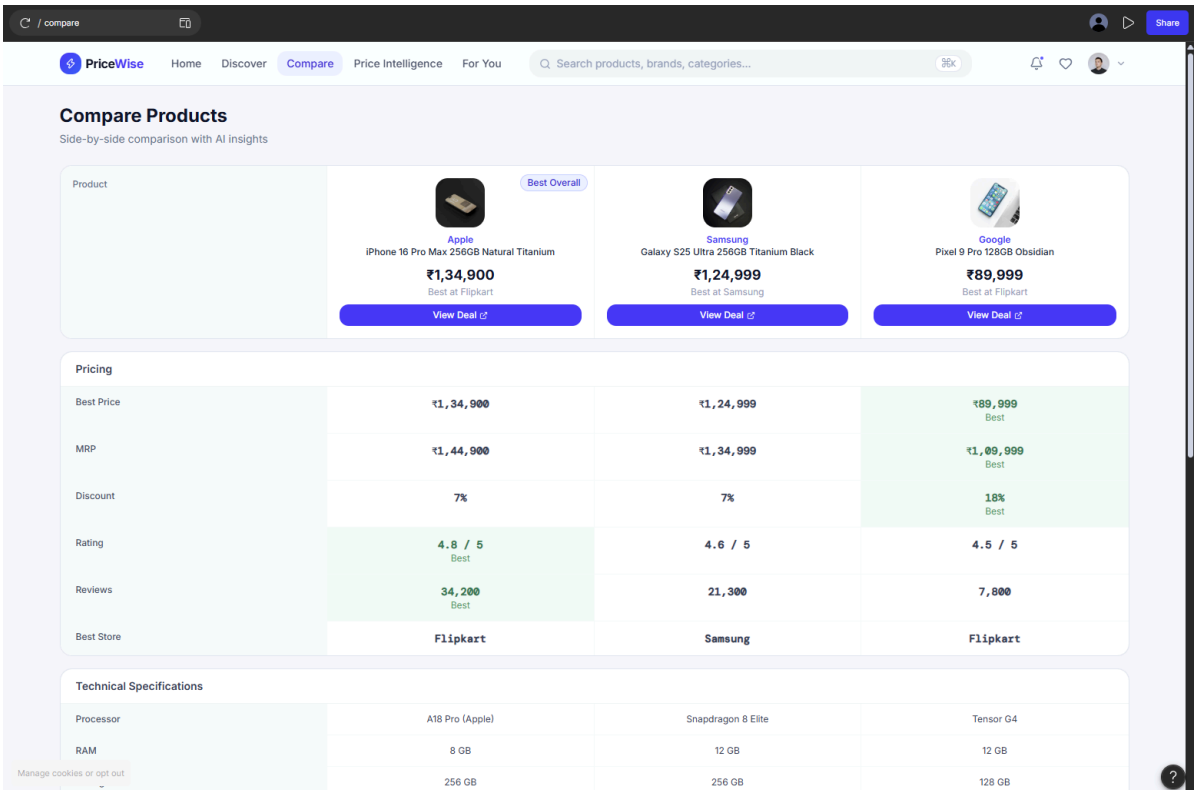


Figure 6.4(b): High-Fidelity Figma UI Mockup — Side-by-Side Product Comparison Matrix

C. Interface Component & Layout Breakdown:

UI Component / Region	Visual Design & Functional Specification
Product Header Row	Crossed-box product image thumbnails, 'Best Value' ribbon badge, product model titles, current best price tags, and primary 'View Deal' external link buttons.
Pricing Breakdown Section	Comparative attribute rows for Best Price, Original MRP, Percentage Discount, User Rating, Review Counts, and Best Store Merchant.
Technical Specifications Matrix	Row-by-row hardware specification breakdown comparing Processors, RAM Memory, Internal Storage, Display Size & Tech, and Battery / Charging specs across all selected items.

D. Screen-Specific Acceptance Criteria:

- Matrix automatically highlights winning attributes in soft green shading with a 'Best' pill badge.
- 'View Deal' CTA buttons open the retailer's verified product page in a new secure browser tab.
- Users can dynamically remove or add additional products to the comparison table.
- Table header row remains sticky when scrolling vertically down long specification sheets.

6.5: Price Intelligence & Predictive Analytics Dashboard

Route: /price-intelligence (Analytics & History)

Screen Purpose & User Goal: Delivers advanced price analytics, historical trendlines, market volatility metrics, record lows, and AI-powered 7/30/60-day price forecasting to ensure users never buy right before a major price drop.

A. Low-Fidelity Blueprint Wireframe (Structural Layout & Placeholders):

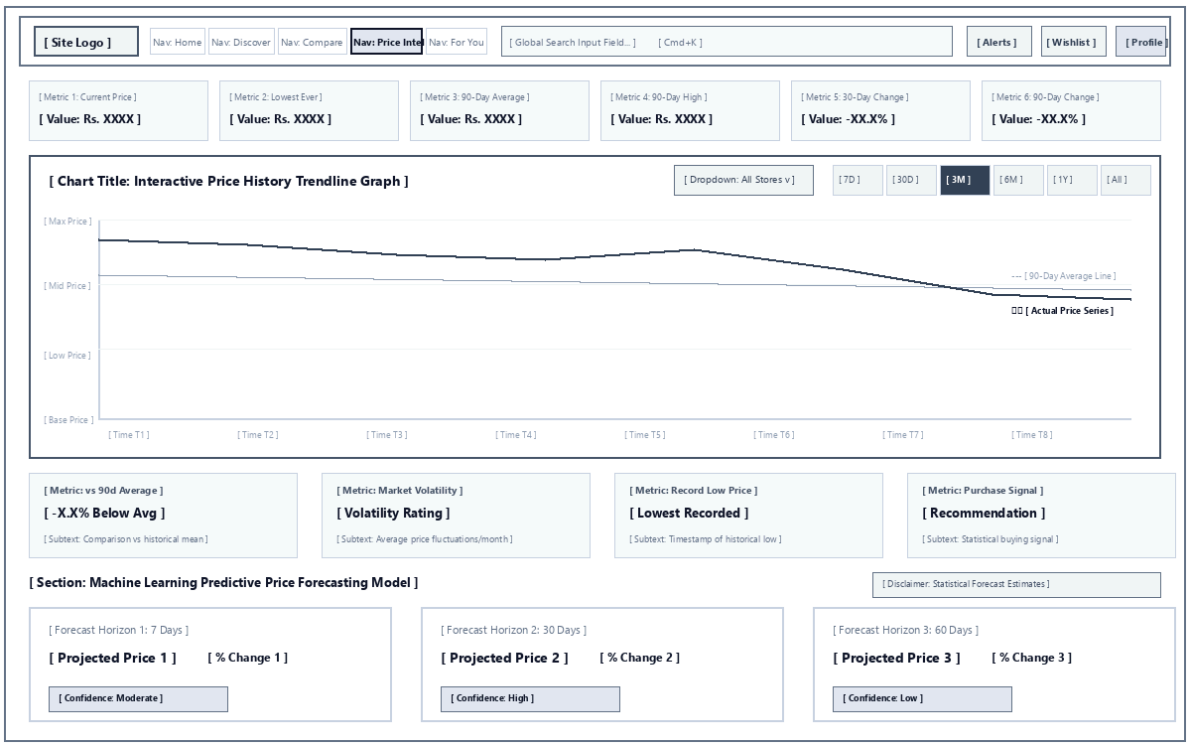


Figure 6.5(a): Low-Fidelity Blueprint Wireframe — Price Intelligence & Predictive Analytics Dashboard

B. High-Fidelity UI/UX Figma Mockup :

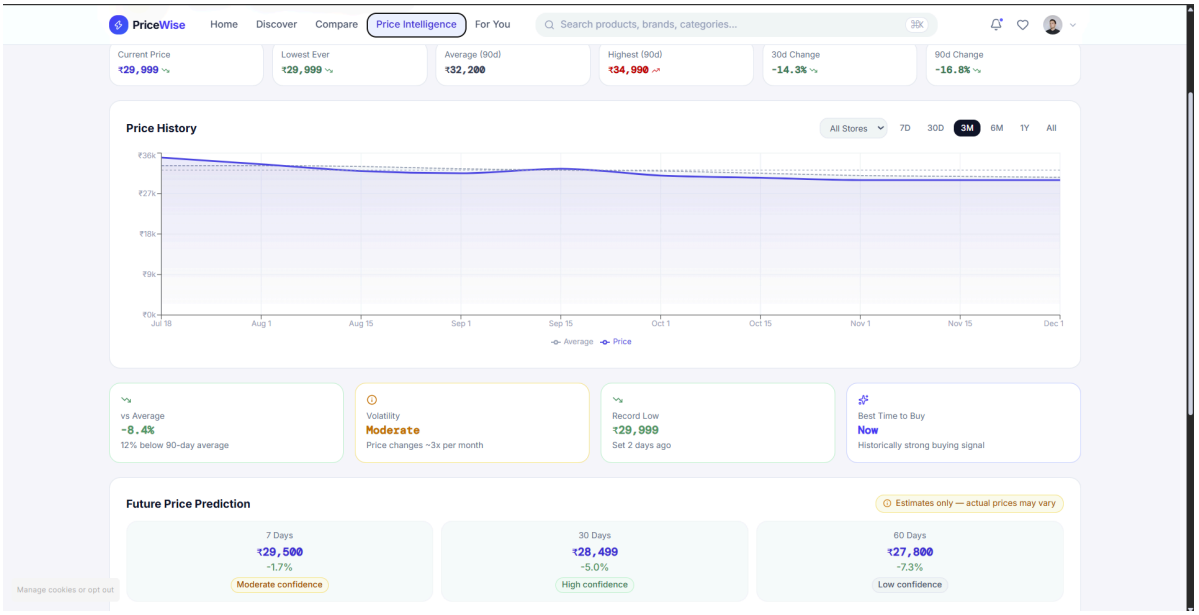


Figure 6.5(b): High-Fidelity Figma UI Mockup — Price Intelligence & Predictive Analytics Dashboard

C. Interface Component & Layout Breakdown:

UI Component / Region	Visual Design & Functional Specification
Top Key Metrics Ribbon	6 summary stat tiles: Current Price, Lowest Ever Recorded, 90-Day Average, 90-Day High, 30-Day Net Change, and 90-Day Net Change.
Interactive Price History Chart	Coordinate axis time-series line chart with dual data series (Actual Price vs 90-Day Average line), store filter dropdown, and timeframe filter pill buttons (7D, 30D, 3M, 6M, 1Y, All).
Market Dynamics Summary Cards	4 analytic insight cards: vs 90-Day Average percentage, Market Volatility score, Record Low timestamp, and Best Time to Buy indicator.
Predictive Price Forecast Horizon	3 machine learning forecast cards projecting 7-Day, 30-Day, and 60-Day price trends with confidence scoring tags and disclaimer note.

D. Screen-Specific Acceptance Criteria:

- Toggling timeframe filters smoothly updates chart coordinate axes and plotted points.
- Hovering over data points displays an interactive tooltip with exact date, store name, and price.
- Confidence badges dynamically adapt color based on certainty (Green = High, Amber = Moderate, Gray = Low).
- Disclaimer banner clearly informs users that future predictions are statistical estimates.

6.6 : Personalized AI Recommendations ('For You')

Route: /for-you (Personalized Recommendations)

Screen Purpose & User Goal: A personalized product discovery feed that analyzes user interaction history, wishlist entries, and search patterns to recommend tailored deals with explicit rationale banners.

A. Low-Fidelity Blueprint Wireframe (Structural Layout & Placeholders):

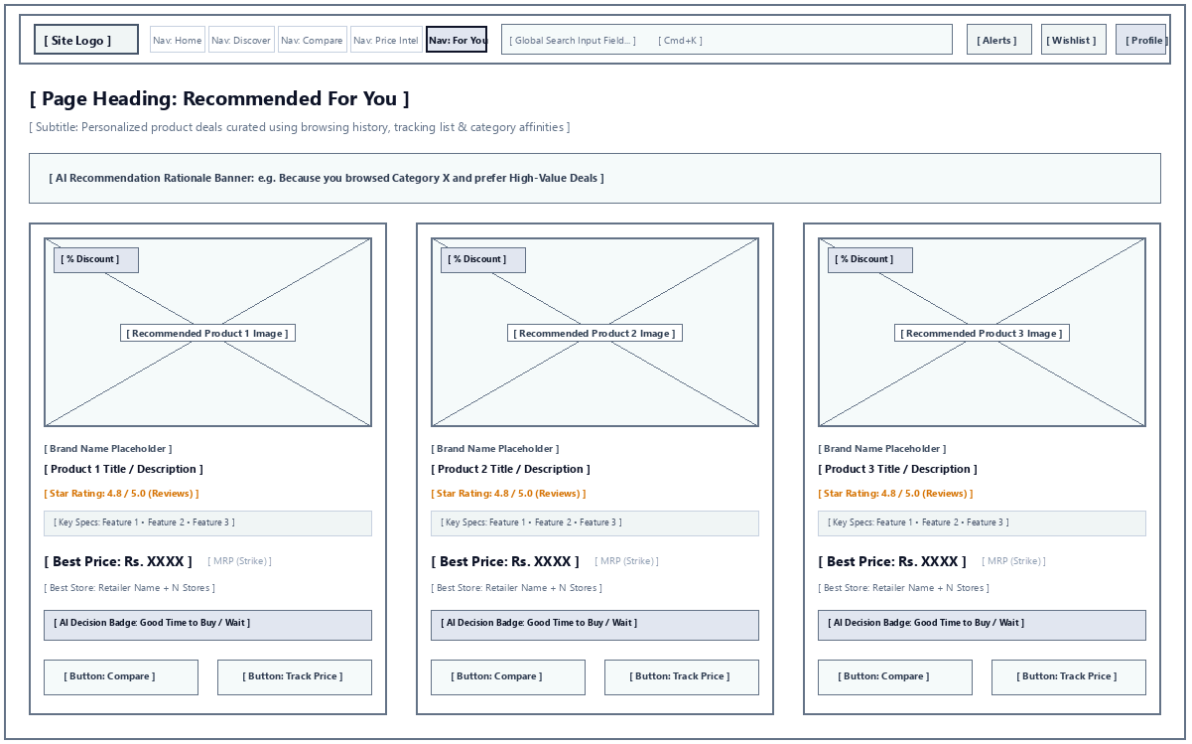


Figure 6.6(a): Low-Fidelity Blueprint Wireframe — Personalized AI Recommendations ('For You')

B. High-Fidelity UI/UX Figma Mockup :

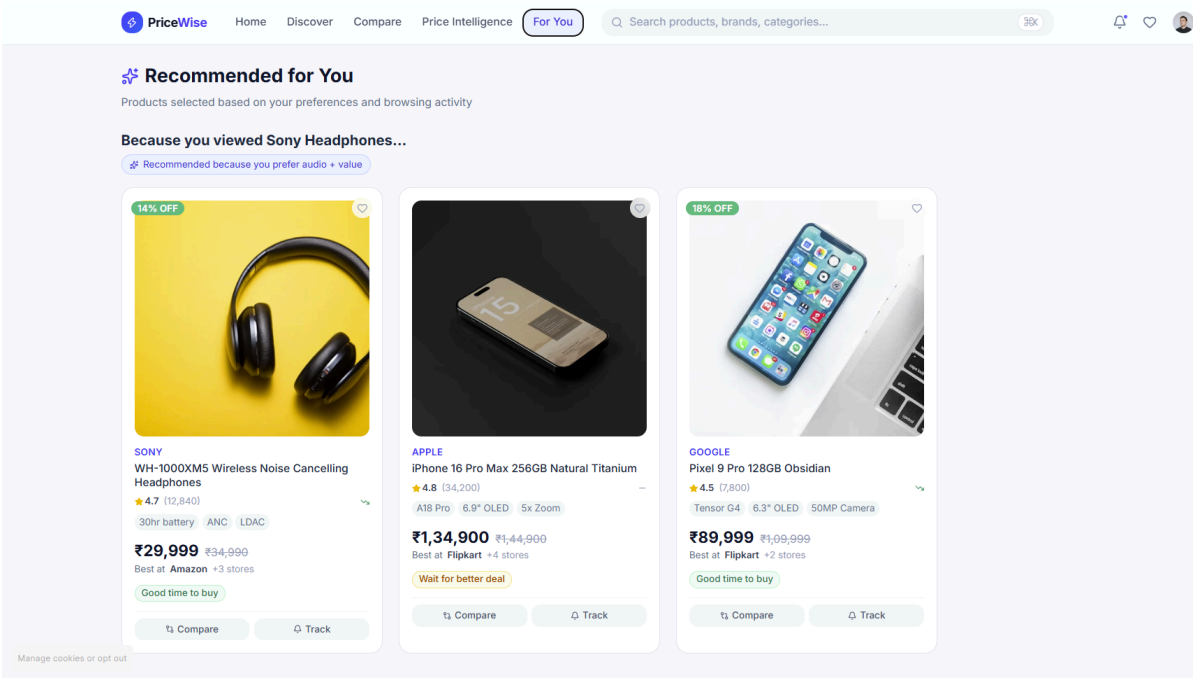


Figure 6.6(b): High-Fidelity Figma UI Mockup — Personalized AI Recommendations ('For You')

C. Interface Component & Layout Breakdown:

UI Component / Region	Visual Design & Functional Specification
Header & Rationale Banner	Section title ('Recommended for You'), descriptive subtitle, and AI recommendation context banner explaining why products were selected (e.g. 'Because you viewed Audio Category...').
Curated Product Cards	Product cards with crossed-box image placeholders, discount badge, brand tag, title, rating, key hardware spec tags, best price vs MRP, best store tag, and AI decision badge.
Quick Action CTAs	Secondary action buttons for 'Compare' and 'Track Price' on each recommended item.

D. Screen-Specific Acceptance Criteria:

- Recommendations refresh dynamically as user browses, tracks, or favorites different categories.
- Rationale banner accurately reflects user's category browsing affinities and price sensitivity.
- Users can dismiss irrelevant recommendations to train the personalization engine.

6.7: Tracked Products & Watchlist Management

Route: /watchlist (Price Tracking)

Screen Purpose & User Goal: Enables users to manage active price tracking targets, monitor price gaps in real-time, view visual progress toward desired thresholds, and modify or pause alert triggers

.

A. Low-Fidelity Blueprint Wireframe (Structural Layout & Placeholders):

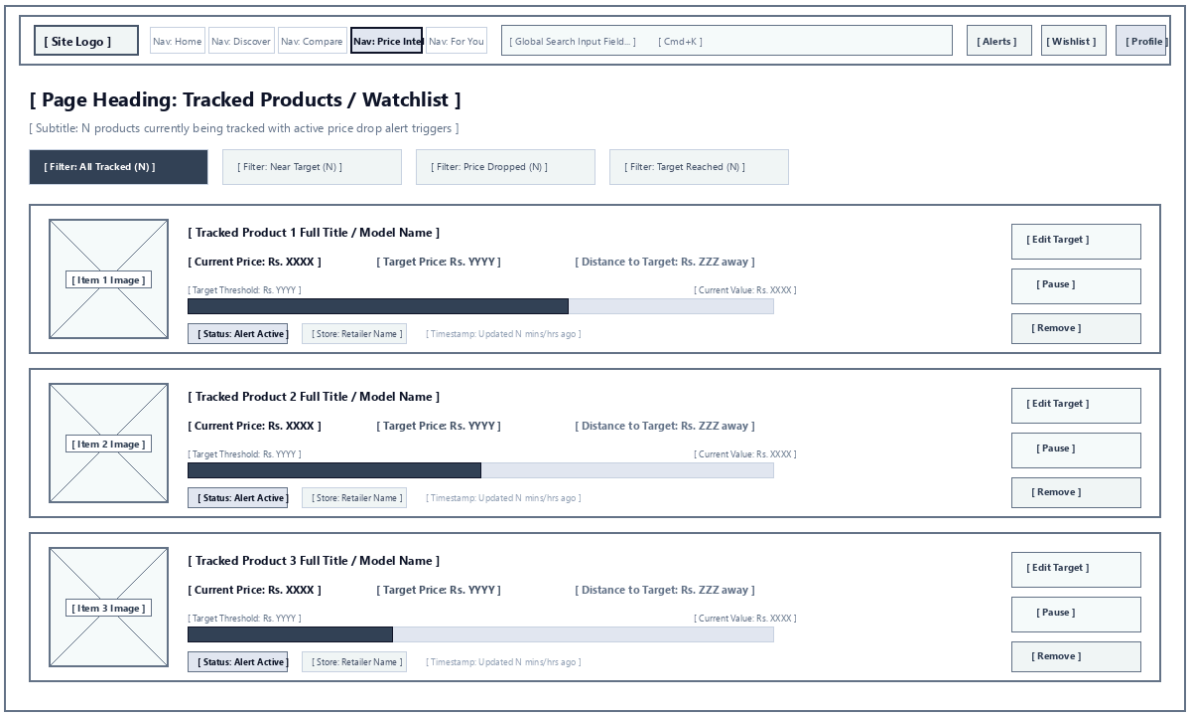


Figure 6.7(a): Low-Fidelity Blueprint Wireframe — Tracked Products & Watchlist Management

B. High-Fidelity UI/UX Figma Mockup :

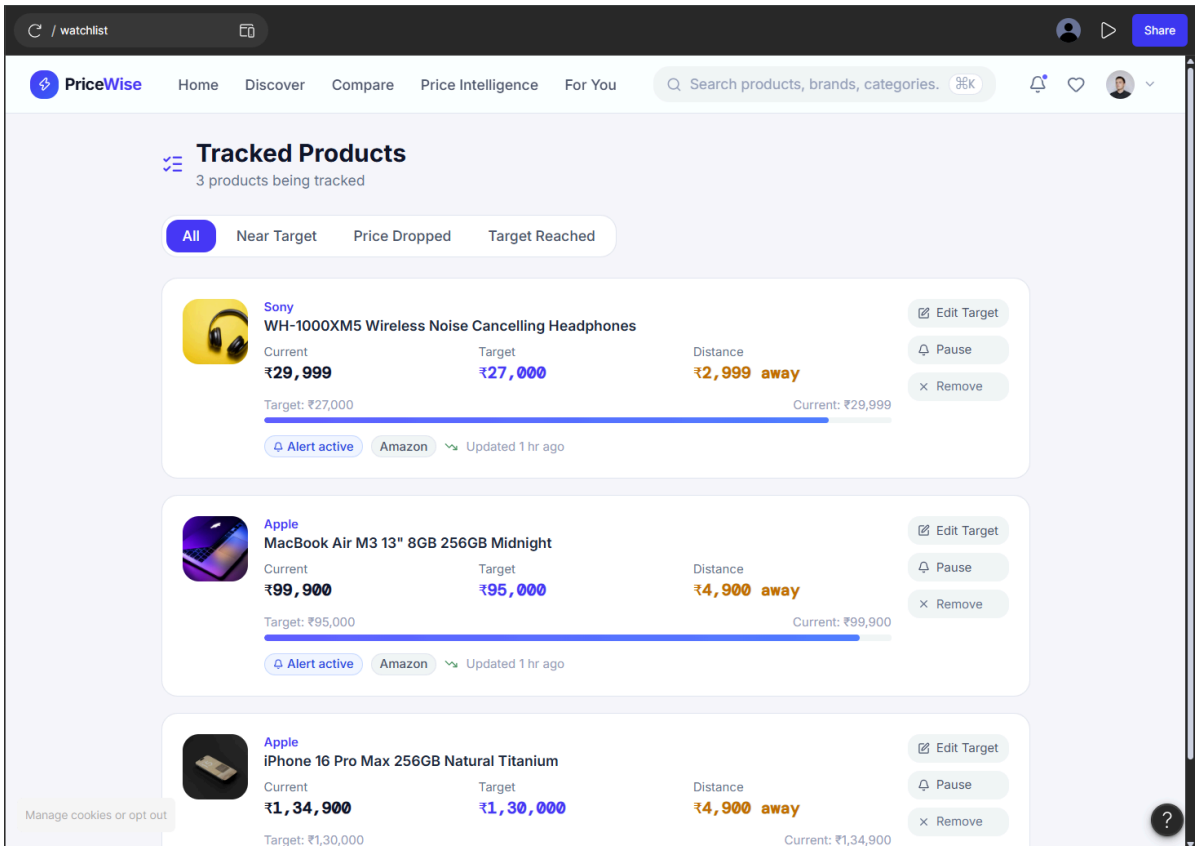


Figure 6.7(b): High-Fidelity Figma UI Mockup — Tracked Products & Watchlist Management

C. Interface Component & Layout Breakdown:

UI Component / Region	Visual Design & Functional Specification
Header & Filter Tabs	Tracked count header ('Tracked Products - N items') and status filter tabs (All, Near Target, Price Dropped, Target Reached).
Tracked Product Card Rows	Horizontal cards with product image placeholder, title, Current Price, Target Price, Distance to Target, and last updated timestamp.
Visual Progress Bar	Horizontal progress bar visualizing current price proximity to user's designated target price threshold.
Alert Status & Actions	Alert active badge, retailer store tag, and right-aligned action buttons: 'Edit Target', 'Pause', and 'Remove'.

D. Screen-Specific Acceptance Criteria:

- Progress bar fills dynamically from left to right as current price approaches designated target price.
- Clicking 'Edit Target' opens an inline input or modal to adjust threshold value.
- Clicking 'Pause' toggles tracking state and updates status badge to 'Alert Paused'.
- Filter tabs accurately filter items based on target proximity and alert status.

6.8: User Wishlist & Saved Products

📍 **Route: /wishlist (Saved Items)**

Screen Purpose & User Goal: A consolidated bookmarking interface displaying all saved products, potential cost savings, and historical price movements since the user added them to their wishlist.

A. Low-Fidelity Blueprint Wireframe (Structural Layout & Placeholders):

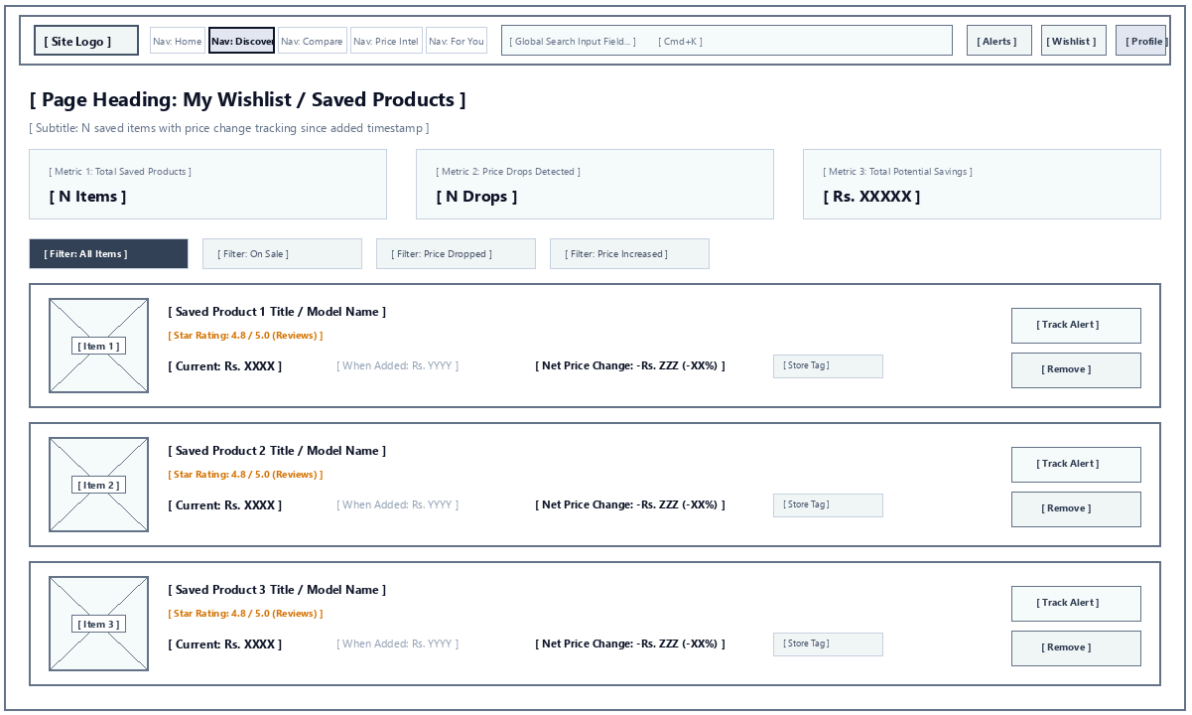


Figure 6.8(a): Low-Fidelity Blueprint Wireframe — User Wishlist & Saved Products

B. High-Fidelity UI/UX Figma Mockup :

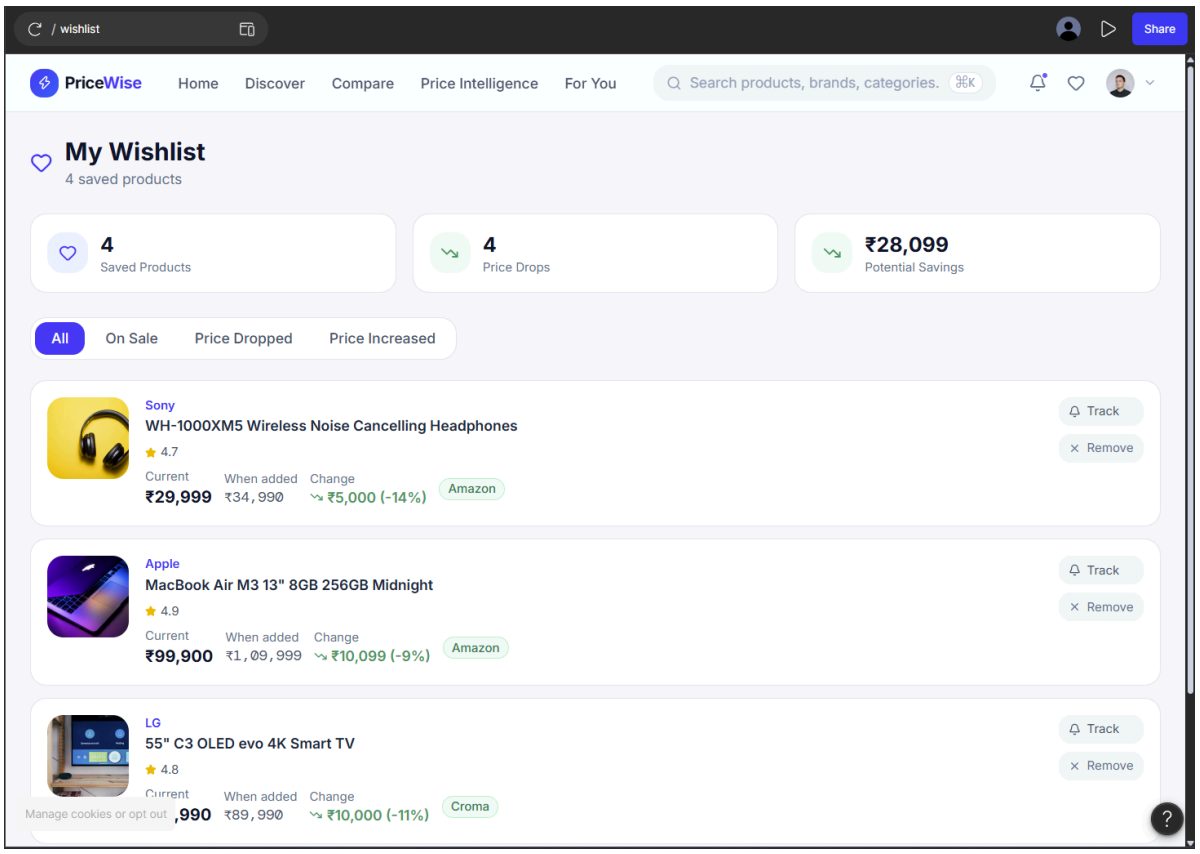


Figure 6.8(b): High-Fidelity Figma UI Mockup — User Wishlist & Saved Products

C. Interface Component & Layout Breakdown:

UI Component / Region	Visual Design & Functional Specification
Wishlist Summary Metrics	3 overview summary tiles: Saved Products Count, Price Drops Detected, and Total Potential Savings.
Filter Bar	Filter buttons for All, On Sale, Price Dropped, and Price Increased.
Saved Product Item Rows	Horizontal item cards with thumbnail image placeholder, title, star rating, Current Price, When Added Price, Net Price Change badge, and store tag.
Row Action CTAs	Secondary action buttons for 'Track' (convert wishlist item to active price alert) and 'Remove' (delete from wishlist).

D. Screen-Specific Acceptance Criteria:

- Total potential savings metric recalculates automatically when items are added or deleted.
- Negative price changes are highlighted in bold green; price increases are highlighted in red.
- Clicking 'Track' converts wishlist item into an active watchlist item with default target price.
- Deleting an item prompts a confirmation toast notification with an 'Undo' option.

6.9: Price Drop Alerts & Notification Center

Route: /alerts (Notifications)

Screen Purpose & User Goal: The notification inbox where users review price drop events, target-reached milestones, and direct merchant checkout links for discounted items.

A. Low-Fidelity Blueprint Wireframe (Structural Layout & Placeholders):

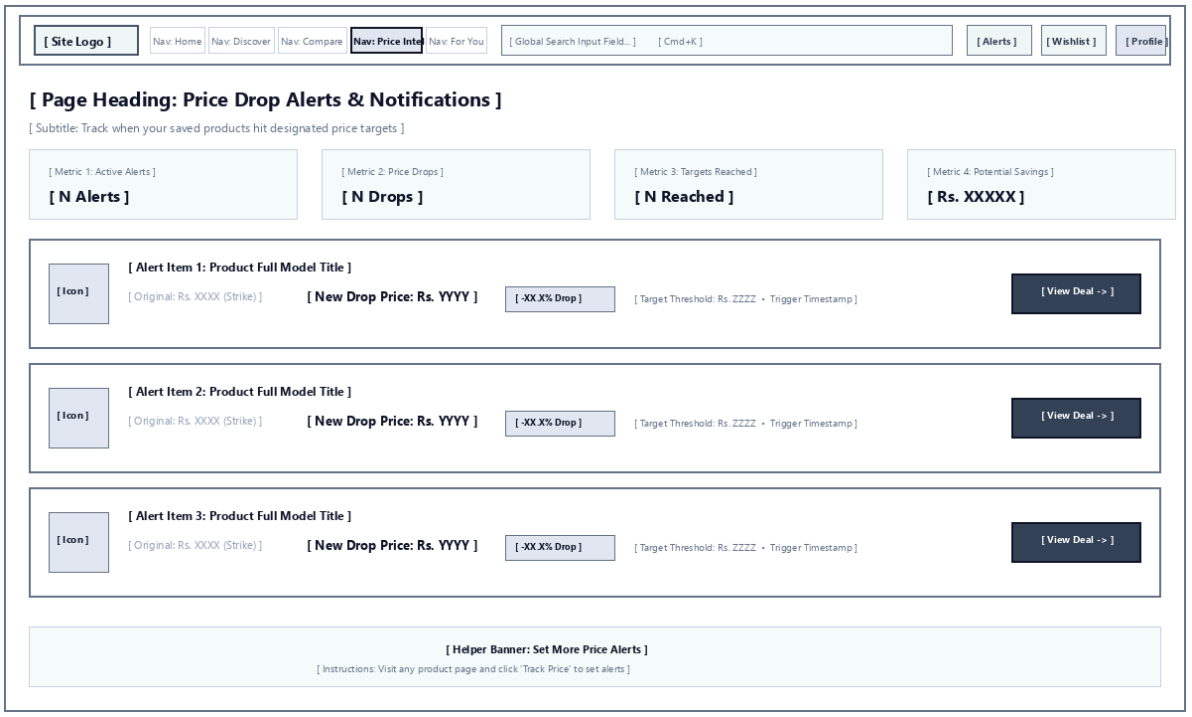


Figure 6.9(a): Low-Fidelity Blueprint Wireframe — Price Drop Alerts & Notification Center

B. High-Fidelity UI/UX Figma Mockup :

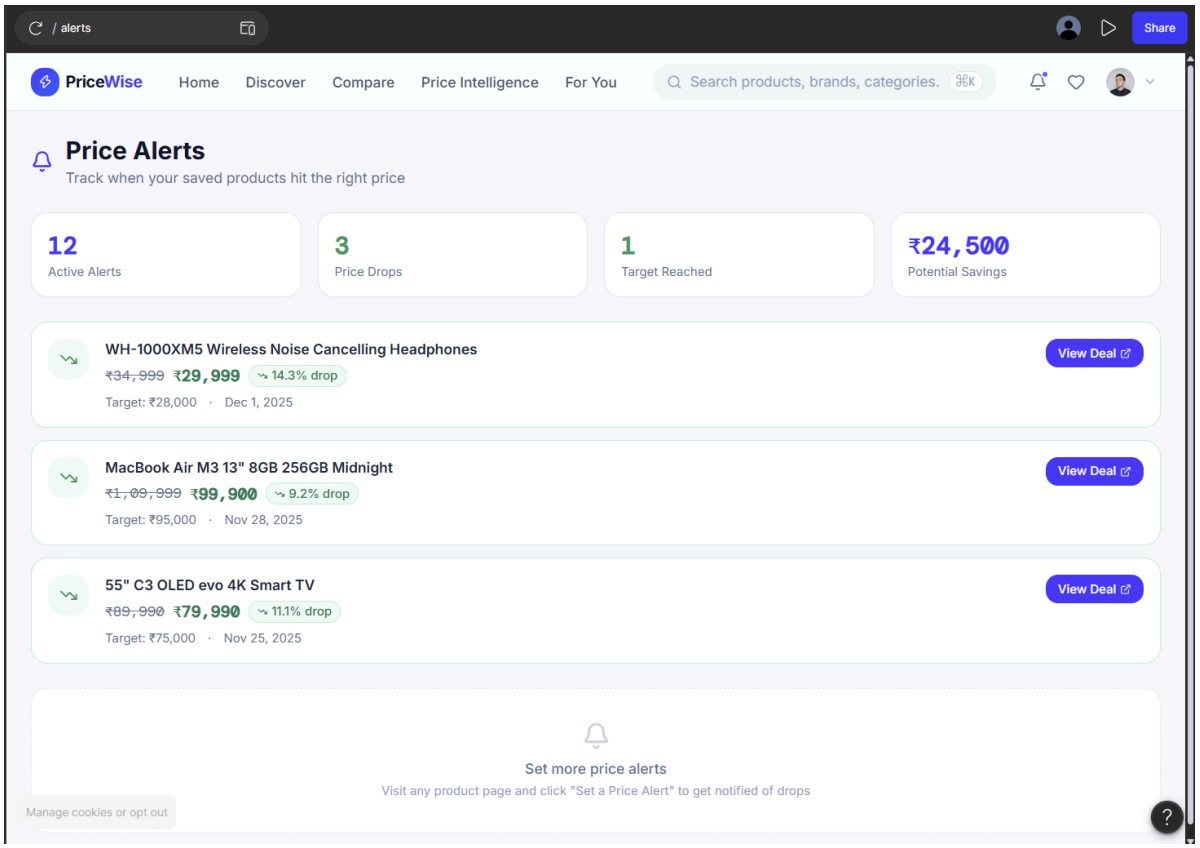


Figure 6.9(b): High-Fidelity Figma UI Mockup — Price Drop Alerts & Notification Center

C. Interface Component & Layout Breakdown:

UI Component / Region	Visual Design & Functional Specification
Alerts Overview Metrics	4 summary metric tiles: Active Alerts, Price Drops, Targets Reached, and Cumulative Potential Savings.
Notification List Rows	Notification cards with price drop icon, product model title, original crossed-out price, new discounted price, percentage drop badge, target price reference, and timestamp.
Direct Checkout CTA	Prominent 'View Deal' button with external link icon opening retailer offer page.
Helper Footer Banner	Subtle helper banner explaining how to configure additional price drop alerts across the catalog.

D. Screen-Specific Acceptance Criteria:

- Unread price drop notifications display an active dot indicator.
- Clicking 'View Deal' routes user through affiliate tracking directly to the retailer product page.
- Notifications can be filtered by 'Price Drops' and 'Targets Reached'.
- Alert timestamps reflect accurate date and time of price drop occurrence.

6.10 : User Profile & Savings Analytics Hub

Route: /profile (Account & Settings)

Screen Purpose & User Goal: Provides user profile overview, lifetime savings metrics, navigation shortcuts to all platform modules, and quick access to recently viewed products.

A. Low-Fidelity Blueprint Wireframe (Structural Layout & Placeholders):

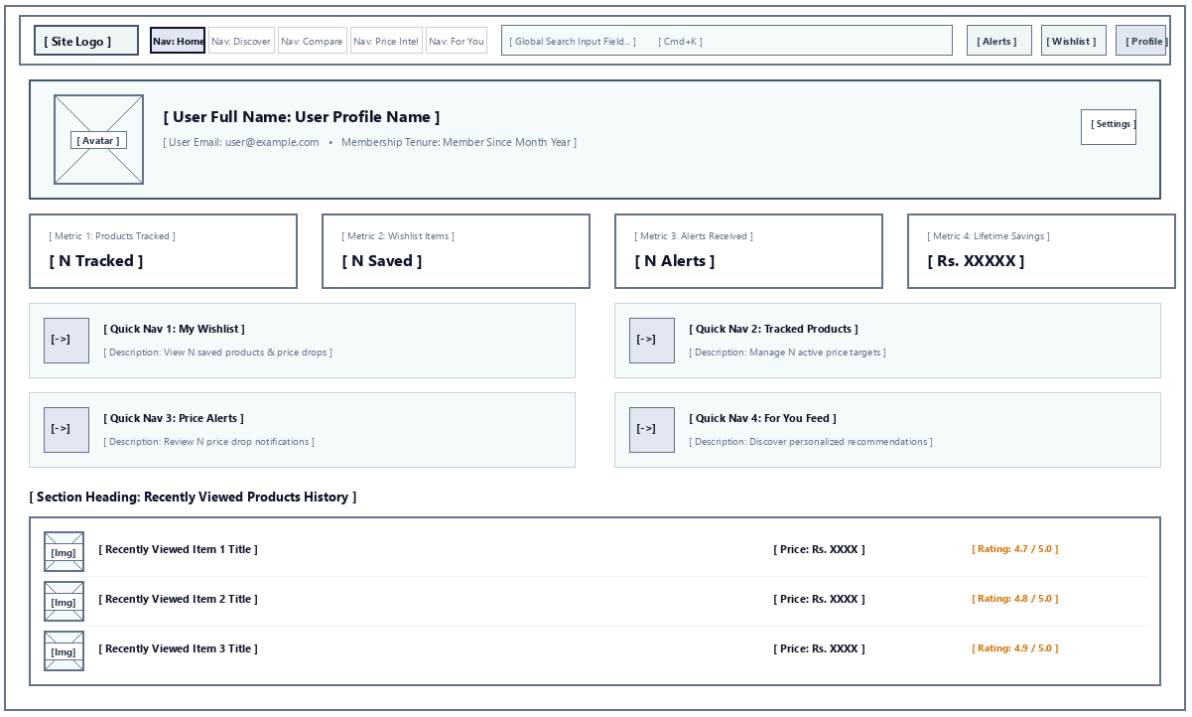


Figure 6.10(a): Low-Fidelity Blueprint Wireframe — User Profile & Savings Analytics Hub

B. High-Fidelity UI/UX Figma Mockup :

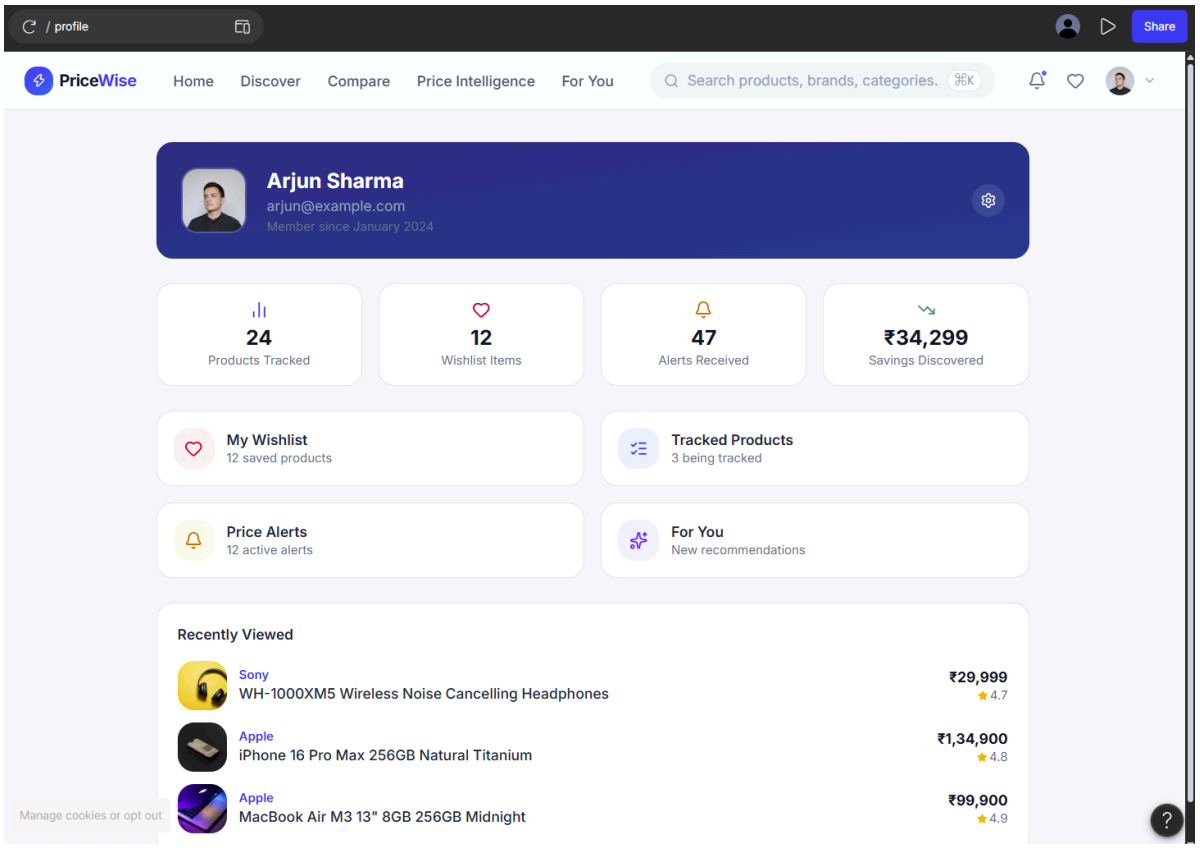


Figure 6.10(b): High-Fidelity Figma UI Mockup — User Profile & Savings Analytics Hub

C. Interface Component & Layout Breakdown:

UI Component / Region	Visual Design & Functional Specification
User Profile Hero Banner	User avatar placeholder, user full name, email address, membership tenure timestamp, and settings configuration button.
Lifetime Key Metrics Grid	4 primary metric cards: Products Tracked, Wishlist Items, Alerts Received, and Total Lifetime Savings Discovered.
Quick Navigation Shortcut Cards	2x2 grid of shortcut cards: My Wishlist, Tracked Products, Price Alerts, and For You Recommendations.
Recently Viewed Products Feed	Compact product history feed showing recently viewed item thumbnails, titles, current prices, and star ratings.

D. Screen-Specific Acceptance Criteria:

- Settings button opens user profile preferences, notification channel toggles, and password management.
- Quick access shortcut cards navigate directly to respective application routes.
- Recently viewed feed maintains a rolling local history of up to 10 most recent items.
- Lifetime savings discovered metric aggregates confirmed discounts below baseline added prices.

7. Acceptance Criteria

The following acceptance criteria define the conditions that must be satisfied for the major PriceWise functionalities to be considered successfully implemented.

ID	Feature	Acceptance Criteria
AC-01	User Authentication	Users can register, log in, and log out successfully. Invalid credentials and incomplete registration details are rejected with appropriate messages.
AC-02	Product Search	Users can search for products using keywords and receive relevant results. Appropriate feedback is displayed when no results are found.
AC-03	Product Details	Users can view product information including name, brand, specifications, image, price, and availability where applicable.
AC-04	Price Comparison	The system displays available prices from supported sources and clearly identifies the lowest available price.
AC-05	Product Comparison	Users can select multiple products and compare their prices, specifications, and key features side-by-side.
AC-06	Price History	Users can view historical price information and a graphical representation of price changes over time when sufficient data is available.
AC-07	Price Prediction	When sufficient historical data exists, the system generates a predicted price/trend and clearly identifies it as an estimate.
AC-08	Buy/Wait/Monitor	The system generates a recommendation based on available price history, current price, and prediction data, with supporting factors where available.
AC-09	Watchlist	Authenticated users can add, view, and remove products from their personal watchlist.

AC-10	Price Alerts	Users can set a target price and receive an alert when the monitored product reaches the specified threshold.
AC-11	Dashboard	Authenticated users can view their watchlist, tracked products, alerts, and relevant price information from the dashboard.
AC-12	Administration	Authorized administrators can manage users, products, and relevant system data, while unauthorized users are denied access.
AC-13	Security	User authentication data is protected, unauthorized access is prevented, and sensitive information is not exposed.
AC-14	Error Handling	The system provides clear and meaningful error messages when an operation fails or invalid information is entered.
AC-15	Responsiveness	The PriceWise interface remains usable and readable across supported desktop and mobile screen sizes.